EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 2 – Lecture 1

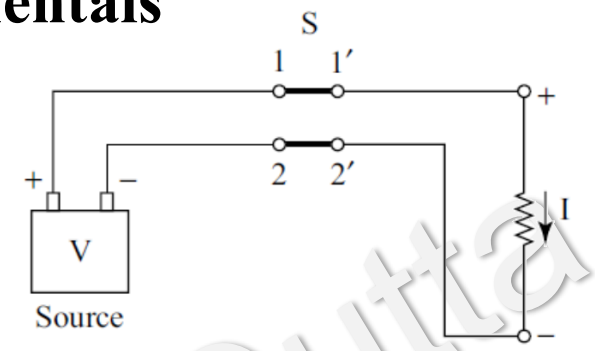
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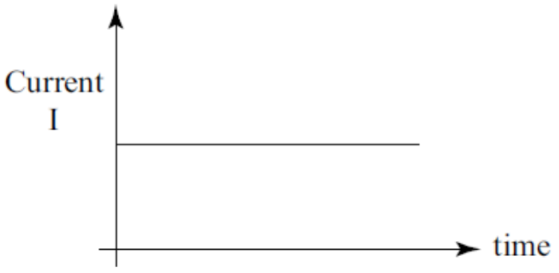
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AC Fundamentals

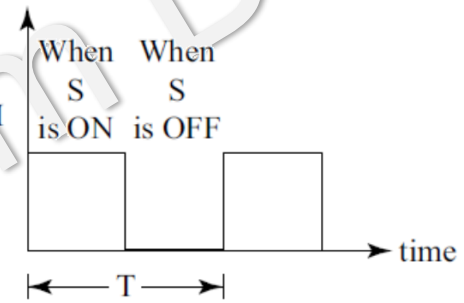
- Direction of current continuously changes
- Polarities of the AC supply source changes alternately and causes AC to flow through the load
- Fig. (a): DC flowing through the load when connected across the battery terminals
- Fig. (b): magnitude is constant
- Fig. (c): nature of current when switch is turned 'ON' and 'OFF' at regular constant intervals
- Fig. (d): using a reverse switching arrangement, we can have an ac flowing through the load
- Supply terminal 1 is connected to load terminal 2′ and supply terminal 2 is connected to load terminal 1′
- If period of switching ON in both directions is kept constant, load current will be alternating and its wave shape will be square or rectangular [Fig. (e) and (f)]
- AC on a large scale is made available by using ac generators



(a)

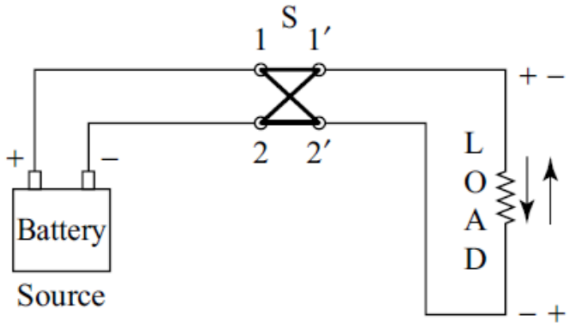


(b)

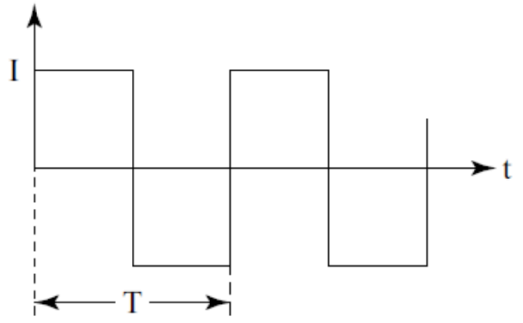


T is called one time period.

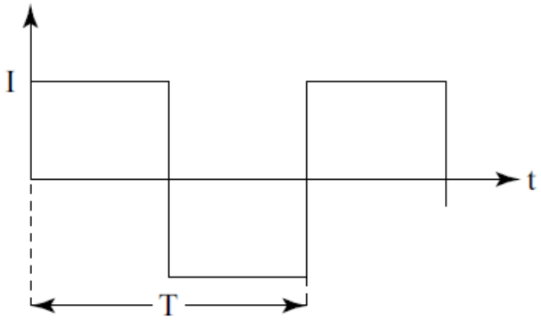
(c)



(d)



(e)



(f)

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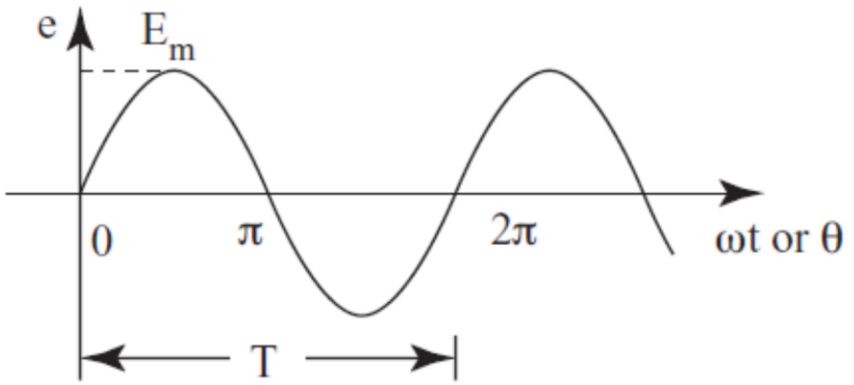
Frequency, Cycle, Time Period, Instantaneous Value, Maximum Value

- Cycle: one set of positive values and the subsequent one set of negative values of an alternating quantity
- Time period (T): time taken for the generation of one cycle of EMF or flow of current caused due to such an EMF
- Frequency (f): total number of cycles of EMF or current produced per second
- The relationship between T and f :

  In T seconds the cycle produced is 1
  In 1 second the cycle produced is 1/T
  Since f is the cycle produced per second, **$f = 1/T$**

- Instantaneous value: the value of an alternating quantity at any instant of time
- For sinusoidal waves, **$e = E_m \sin\theta$** and **$i = I_m \sin\theta$**
- At $\theta = 0°$, $e = E_m \sin 0° = 0$
- At $\theta = 90°$, $e = E_m \sin 90° = E_m$

  E_m: maximum value which occurs at $\theta = 90°$



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Average Value

- Sum of all its values divided by the total number of values
- The pattern of wave repeats after every cycle
- The sum of all the values in a cycle is the integration of its values over a period of time
- A full cycle is formed in 2π radians or in T seconds
- A symmetrical wave is one where the positive half cycle is exactly the same as the negative half cycle. If we integrate the values for a complete cycle and take its average over one cycle, the quantity becomes equal to zero. The average value if calculated over a complete cycle would become zero

$$V_{av} = \frac{1}{2\pi}\int_0^{2\pi} v\, d\theta = 0 \qquad V_{av} = \frac{1}{T}\int_0^{T} v\, dt = 0$$

- Average value of a sinewave or any other symmetrical wave over a complete cycle is zero
- When we intend to calculate the average value of such waves, we calculate the average value for one-half cycle

$$V_{av} = \frac{1}{\pi}\int_0^{\pi} v\, d\theta \qquad V_{av} = \frac{2}{T}\int_0^{T/2} v\, dt$$

- The average value of an alternating voltage or current is related to the peak voltage or current as:

$$V_{av} = \frac{2\,V_m}{\pi} = 0.637\,V_m \qquad I_{av} = \frac{2\,I_m}{\pi} = 0.637\,I_m$$

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Effective or RMS Value

- The effective value or RMS value of an alternating quantity is determined by considering equivalent heating effect
- The effective value of an alternating quantity (say current) is that value of dc current which when flowing through a given circuit element (say a resistance element) for a given time will produce the same amount of heat as produced by the alternating current when flowing through the same circuit element for the same time
- This is equal to the square root of the mean of the squares of instantaneous values of the alternating quantities

$$I = \sqrt{\frac{i_1^2 + i_2^2 + i_3^2 + \cdots + i_n^2}{n}}$$

- The RMS value of an alternating current is same for one full cycle or one half of a cycle:

$$I = \sqrt{\frac{1}{T}\int_0^T i^2 dt} \qquad \text{or} \quad I = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} i^2 \, d\theta}$$

- The RMS value of an alternating current or voltage is related to the maximum or peak current or voltage as:

$$I = \frac{I_m}{\sqrt{2}} = 0.707 \, I_m \qquad V = \frac{V_m}{\sqrt{2}} = 0.707 \, V_m$$

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Form Factor

- **Form factor** is an indicator of the shape or the form of the ac wave.
- It is the ratio of the RMS value to the average value of an alternating quantity.
- For a sinusoidal varying quantity, the form factor K_f is

$$K_f = \frac{\text{RMS value}}{\text{Average value}} = \frac{0.707\ I_m}{0.637\ I_m} = 1.11$$

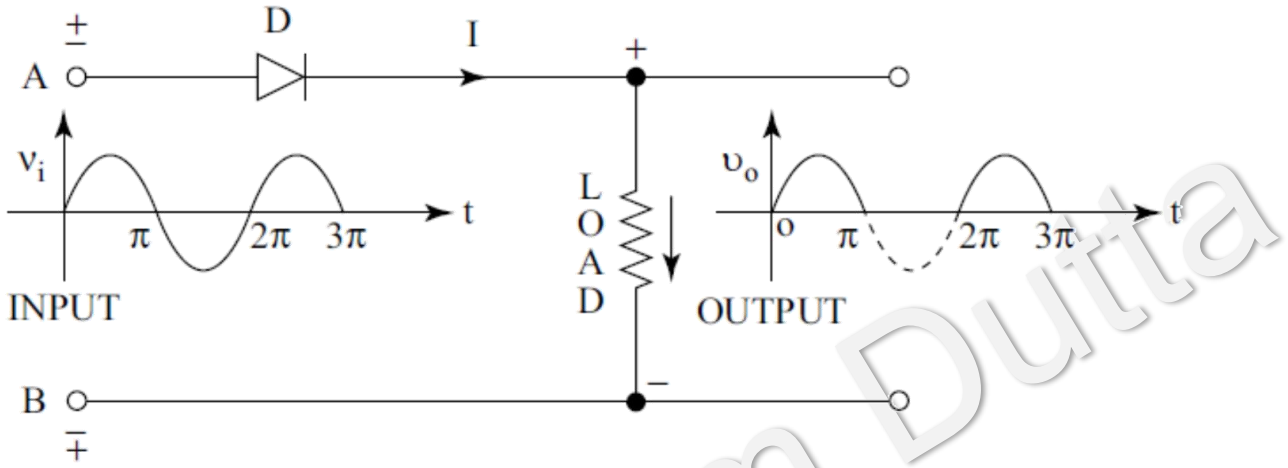
- The sharper the wave shape, the more will be the value of the form factor.

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Peak Factor

- The **peak or crest factor** is the ratio of peak or maximum value to its rms value.

Half-wave-rectified Alternating Quantity



- Half-wave-rectified quantity, whether voltage or current will have its one half cycle blocked by using a diode rectifier.
- Since the diode allows current to flow in one direction only, current through the load resistance will flow, in one direction only.
- During the negative half cycle of the input voltage the diode will block current flow and hence no voltage will be appearing across the load during all negative half cycles.
- For half-wave-rectified current or voltage, we have to consider the current or voltage which is available for the positive half cycles and average it for the complete cycle.
- For a complete cycle, i.e., from 0 to 2π, current flows only from 0 to π.

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RMS and Average Values of Half-wave-rectified Alternating Quantity

- **RMS value:** We have to square the quantity, take its sum from 0 to π and then take the average for the whole cycle, i.e., from 0 to 2π. Thus, the RMS value for a half-wave-rectified current, is calculated as

$$I = \sqrt{\frac{1}{2\pi}\int_0^{\pi} I_m^2 \sin^2\theta \, d\theta} = \frac{I_m}{2}$$

- **Average value:** We take the sum of the quantity from 0 to π and then take the average for the whole cycle, i.e., from 0 to 2π. Thus, the average value for a half-wave-rectified current, is calculated as

$$I_{av} = \frac{1}{2\pi}\int_0^{\pi} i \, d\theta = \frac{1}{2\pi}\int_0^{\pi} I_m \sin\theta \, d\theta = \frac{I_m}{\pi}$$

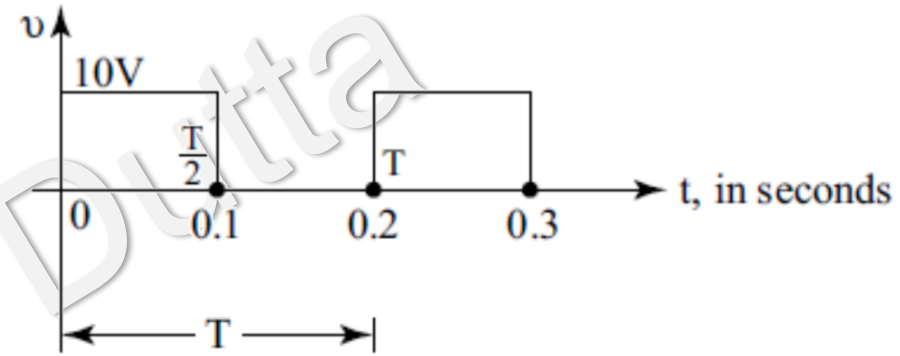
- **Form factor:** $K_f = \frac{\text{RMS value}}{\text{Average value}} = \frac{I_m}{2} \times \frac{\pi}{I_m}$

$$= \frac{\pi}{2} = \frac{3.14}{2} = 1.57$$

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Problem

Q: Calculate the RMS value, average value and form factor of a half-rectified square voltage.



A: $V_{av} = \frac{1}{T}\int_0^{T/2} v\,dt$

$T = 0.2,\ v = 10\ V$

$$V_{av} = \frac{1}{0.2}\int_0^{0.1} 10\,dt = \frac{1}{0.2}[10t]_0^{0.1} = \frac{1}{0.2} \times 10 \times 0.1 = 5\ V$$

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^{T/2} v^2\,dt} = \sqrt{\frac{1}{0.2}\int_0^{0.1} 10^2\,dt}$$

$$= \sqrt{\frac{1}{0.2}\left[[10]^2\,t\right]_0^{0.1}} = \sqrt{\frac{1}{0.2}[100 \times 0.1]}$$

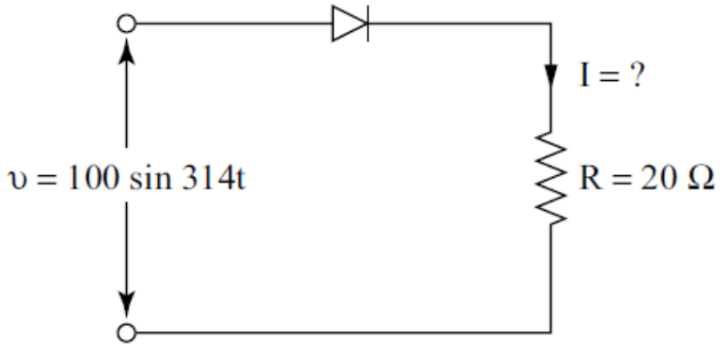
$$= \sqrt{\frac{10}{0.2}} = \sqrt{50} = 7.09\ V$$

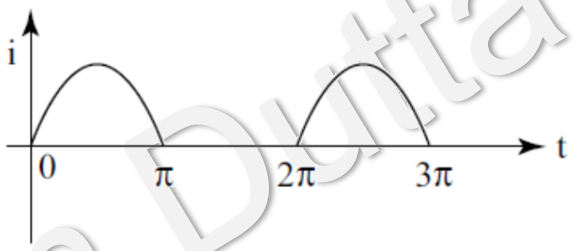
Form factor, $K_f = \frac{V_{rms}}{V_{av}} = \frac{7.09}{5} = 1.4$

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Problem

Q: An alternating voltage of 100 sin 314 t is applied to a half-wave diode rectifier which is in series with a resistance of 20 Ω. What is the RMS value of the current drawn from the supply source?





We have, $v = V_m \sin \omega t = 100 \sin 314\, t$

$V_m = 100$ V

For a full sine wave, RMS value

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

For half-rectified wave

$$V_{rms} = \frac{V_m}{2}$$

Therefore,

$$I_{rms} = \frac{V_{rms}}{R} = \frac{V_m}{2 \times 20} = \frac{100}{2 \times 20} = 2.5\,\text{A}$$

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Problem

Q: An alternating sinusoidal voltage of $\upsilon = 150 \sin 100\,\pi\, t$ is applied to a circuit which offers a resistance of 50 Ω to the current in one direction and prevents the flow of any current in the opposite direction. Calculate the RMS and average values of the current and the form factor. What is the frequency of the supply?

A:

$$v = 150 \sin 100\,\pi t$$

standard form, $v = V_m \sin \omega t$

The maximum value of voltage, $V_m = 150$

$$\omega = 100\,\pi = 2\,\pi f$$

frequency, $f = 50\ \text{Hz}$

The circuit in the question is a half-wave-rectified one.

For half sine wave, the RMS value

$$V_{rms} = \frac{V_m}{2} \text{ and } V_{av} = \frac{V_m}{\pi}$$

Therefore,

$$I_{rms} = \frac{V_{rms}}{R} = \frac{V_m}{2R} = \frac{150}{2 \times 50} = 1.5\ \text{A}$$

$$I_{av} = \frac{V_{av}}{R} = \frac{V_m}{\pi R} = \frac{150}{3.14 \times 50} = 0.95\ \text{A}$$

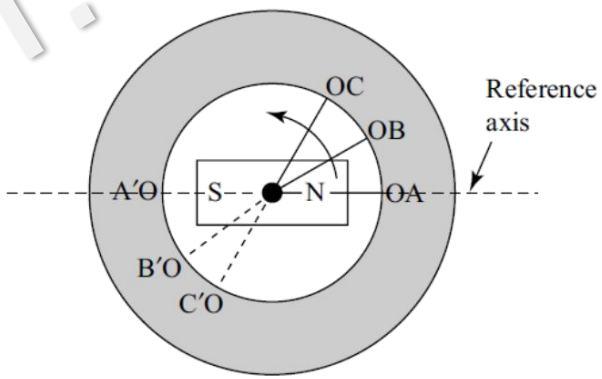
Form factor,

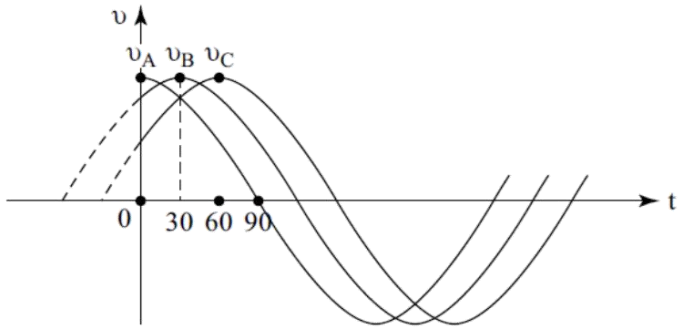
$$K_f = \frac{I_{rms}}{I_{av}} = \frac{1.5}{0.95} = 1.57$$

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Phase and Phase Difference

- The position of a coil or a set of coils forming a winding with respect to some axis of reference is called its phase.
- If three coils are placed at different angles with respect to the reference axis, there exists a space phase difference between these three coils AA′, BB′, and CC′.
- When EMFs will be induced in these coils due to the cutting of the magnetic flux or due to change in flux linkages, the EMFs will have similar time phase difference between them
- A magnet is shown rotating in the anticlockwise direction. Maximum flux will be cut by the coil AA′ at time, t = 0. Hence, maximum voltage will be in the coil AA′ at time, t = 0. Maximum flux will be cut by the coil BB′ after an elapse of 30°, i.e., by the time the rotating magnet rotates by 30°. Similarly, maximum flux will be cut by the coil CC′ after an elapse of time represented by 60°.
- The voltage waves in coil AA′, BB′, and CC′ will, have a time phase difference of 30°. Since voltage υ_A is appearing earlier than υ_B, υ_A is said to be leading than voltage υ_B. Voltage induced in the three coils AA′, BB′, and CC′ will have a time phase difference of 30°.

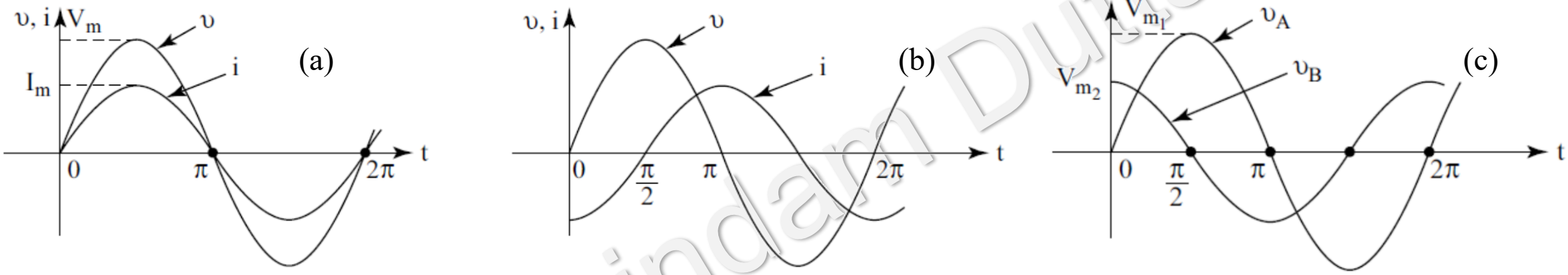




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Phase and Phase Difference

If current in a circuit changes in accordance with the voltage, i.e., when the voltage is at its maximum value, the current is also at its maximum value, and when the voltage starts increasing in the positive direction from its zero value, the current also starts increasing in the positive direction from its zero value; then, the voltage and current are said to be in phase. Note that the magnitudes of voltage and current may be different.



The expressions for voltage and current for Fig. (a) can be written as

$$\upsilon = V_m \sin \omega t$$

$$i = I_m \sin(\omega t + 0) = I_m \sin \omega t$$

The voltage and current shown in Fig. (b) can be represented as

$$v = V_m \sin \omega t$$

$$i = I_m \sin(\omega t - \pi/2)$$

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Phase and Phase Difference

If current is leading the voltage by $\pi/2$ degrees, we will represent the current, i as

$$i = I_m \sin(\omega t + \pi/2)$$

If two voltages ν_A and ν_B are represented as in Fig. (c), they can be expressed as

$$\nu_A = V_{m_1} \sin \omega t$$

$$\nu_B = V_{m_2} \sin(\omega t + \pi/2)$$

This is because voltage ν_B is leading the voltage ν_A. The maximum value of ν_B is appearing $\pi/2$ degrees before the maximum value of ν_A appears. However, if ν_B is taken as the reference voltage we can express ν_B and ν_A as

$$\nu_B = V_m \sin \omega t$$

$$\nu_A = V_m \sin(\omega t - \pi/2)$$

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AC applied across a pure resistor

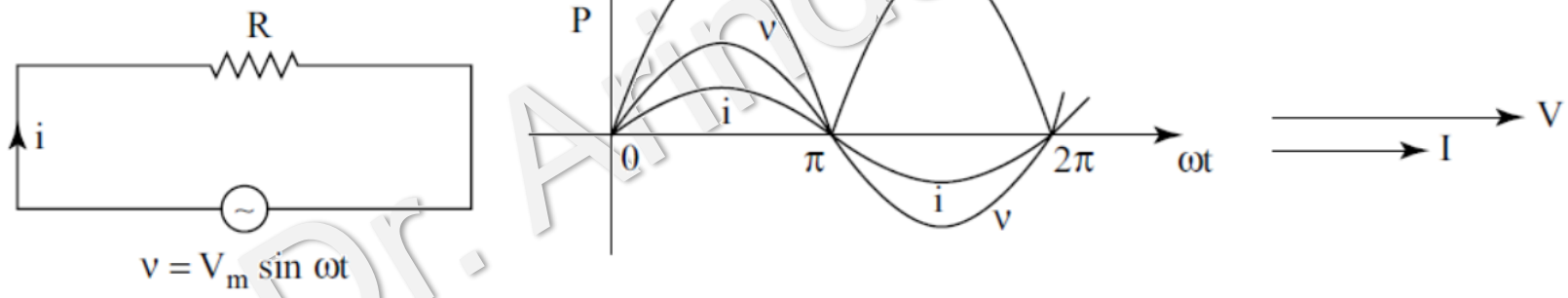
The instantaneous value of

Voltage, v of the source is $v = V_m \sin \omega t$

Where, V_m is the maximum value of the voltage in Volts; $\omega = 2\pi f$ rad/sec; and f is the frequency of supply voltage in cycles per second.

The current flowing though the circuit will be $i = \frac{v}{R} = \frac{V_m \sin \omega t}{R}$

At $\omega t = \frac{\pi}{2}$, $i = \frac{V_m}{R}$



The maximum value of i is I_m

Therefore, $I_m = \frac{V_m}{R}$

Thus, I can be written as

or, $i = \frac{V_m}{R} \sin \omega t$

$i = I_m \sin \omega t$

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AC applied across a pure resistor

Average power in a purely resistive circuit, $P = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} = V I$

Power factor:

Power factor is the cosine of the phase angle between voltage and current.

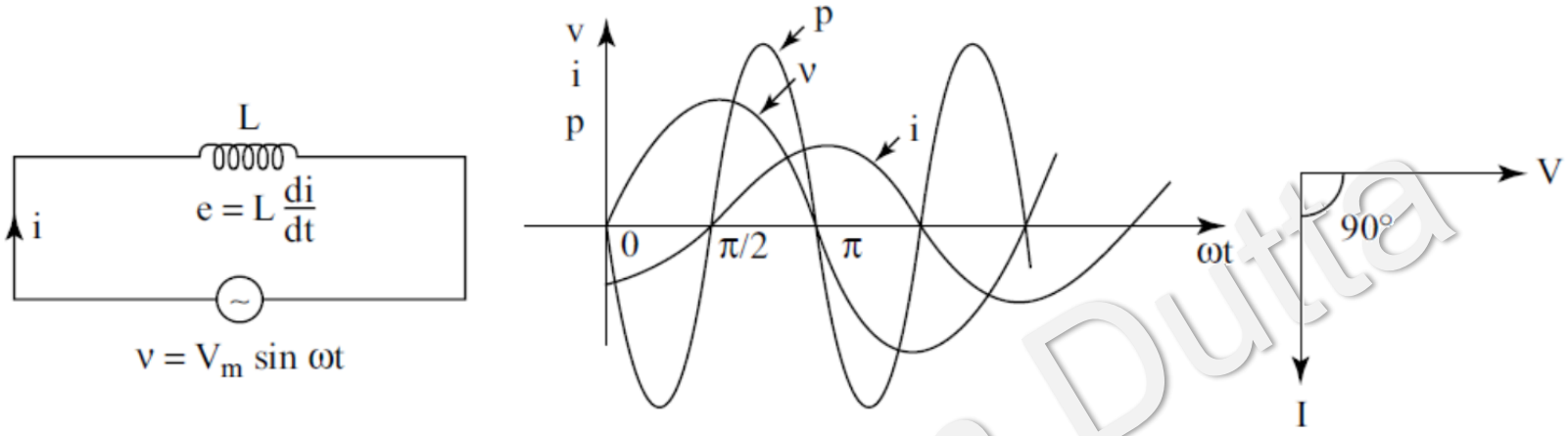
In a resistive circuit the phase difference between voltage and current is zero, i.e., they are in phase.

So the phase angle $\theta = 0$.

Power factor, $P.f = \cos\theta = \cos 0^\circ = 1$

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AC applied across a pure inductor



In a purely inductive circuit: $v = V_m \sin \omega t$

and $i = I_m \sin\left(\omega t - \frac{\pi}{2}\right)$

$= I_m \sin(\omega t - 90°)$

The current, i is also sinusoidal but lagging behind, v by 90°.

Power factor, $\cos\phi = \cos 90° = 0$

Inductor stores energy in one half of cycle and returns it in other half, leading to no net power consumption over a complete cycle.

So, average power absorbed by a pure inductor, P = 0.

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AC applied across a pure inductor

Here $$I_m = \frac{V_m}{\omega L}$$

The opposition to current is ωL. This is called inductive reactance, X_L which is $X_L = \omega L = 2\pi fL\ \Omega$. The opposition offered by an inductor to the flow of current is X_L which is equal to ωL. This is called the inductive reactance and is expressed in Ohms. Inductance, L is expressed in Henry.

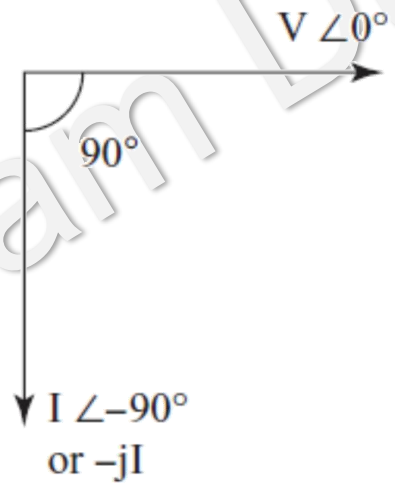
RMS values:

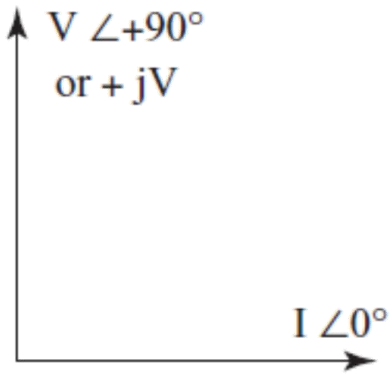
$I_m = \frac{V_m}{\omega L}$ can be written as

$$\frac{I_m}{\sqrt{2}} = \frac{V_m/\sqrt{2}}{\omega L}$$

$$I = \frac{V}{X_L}$$

$$V = I X_L$$



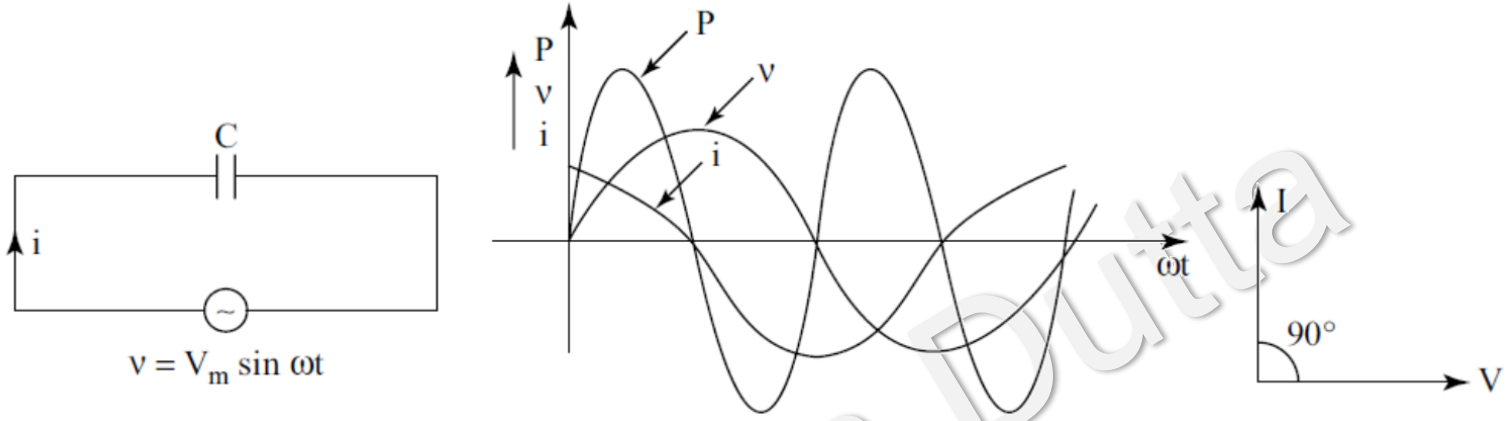


If V is taken as the reference axis, we can represent V as a phasor and represent as V∠0°. Since current, I is lagging voltage, V by 90°, we represent the current as I ∠− 90° or -jI for a purely inductive circuit. Again, if I is taken as the reference axis, then I and V can be represented as I∠0°. And V∠+ 90° or +jV, respectively.

j is an operator which indicates rotation of a phasor by 90° in the **anticlockwise** direction from the reference axis.

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AC applied across a pure capacitor



Hence, in a pure capacitive circuit, $v = V_m \sin \omega t$ and current $i = I_m \sin (\omega t + \frac{\pi}{2})$. Current leads the voltage by 90°.

$X_C = \frac{1}{\omega C}$ is called the capacitive reactance of the capacitor.

To express in terms of RMS values,

We take $\frac{I_m}{\sqrt{2}} = I$ and $\frac{V_m}{\sqrt{2}} = V$

Therefore, $I_m = \frac{V_m}{X_C}$ can be written as

$$I = \frac{V}{X_C}$$

Since current leads voltage by 90°, power factor:

$$P.f = \cos \phi = \cos 90° = 0$$

During one half of the cycle, the capacitor receives and stores energy from the source, and then during the next half of the cycle, it returns that same amount of energy back to the circuit, resulting in no net power consumption over a complete cycle. **So, average P = 0**

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Problem

Q: An alternating voltage of RMS value 100 V, 50 Hz is applied separately across a resistance of 10 Ω, an inductor of 100 mH, and a capacitor of 100 μF. Calculate the current flow in each case. Also draw and explain the phasor diagrams.

A:

$$R = 10\ \Omega$$

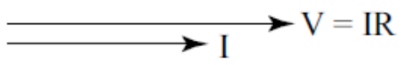
$$X_L = \omega L = 2\pi f L = 2 \times 3.14 \times 50 \times 100 \times 10^{-3}\ \Omega$$

$$= 31.4\ \Omega$$

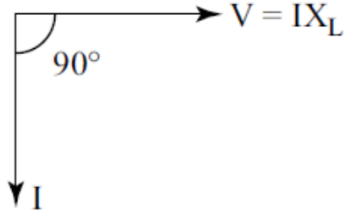
$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2 \times 3.14 \times 50 \times 100 \times 10^{-6}}$$

$$= \frac{10^6}{314 \times 100} = 31.8\ \Omega$$

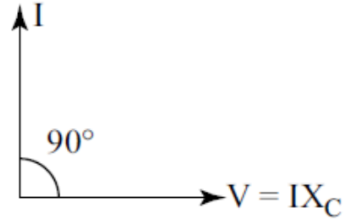
Current through R, $= \frac{100}{10} = 10\ A$



Current through L, $= \frac{100}{X_L} = \frac{100}{31.4} = 3.18\ A$



Current through C, $= \frac{100}{X_C} = \frac{100}{31.8} = 3.1\ A$



In a resistive circuit current is in phase with the applied voltage; in a purely inductive circuit current lags the voltage by 90°; and in a purely capacitive circuit current leads the voltage by 90°.

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Problem

Q: An inductor of 0.5 H is connected across a 230 V, 50 Hz supply. Write the equations for instantaneous values of voltage and current.

A:

$$V = 230\ V,\ V_m = \sqrt{2}\ V = 1.414 \times 230 = 324\ V$$

$$X_L = \omega L = 2\pi\ fL = 2 \times 3.14 \times 50 \times 0.5\ \Omega = 157\ \Omega$$

$$I = \frac{V}{X_L} = \frac{230}{X_L} = \frac{230}{157} = 1.46\ A$$

$$I_m = \sqrt{2}\ I = 1.414 \times 1.46 = 2.06\ A$$

The equations are,

$$V = V_m \sin \omega t = 324 \sin \omega t = 324 \sin 2\pi\ ft = 324 \sin 314t$$

and

$$i = I_m \sin\left(\omega t - \frac{\pi}{2}\right) = 2.06 \sin\left(314t - \frac{\pi}{2}\right)$$

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EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

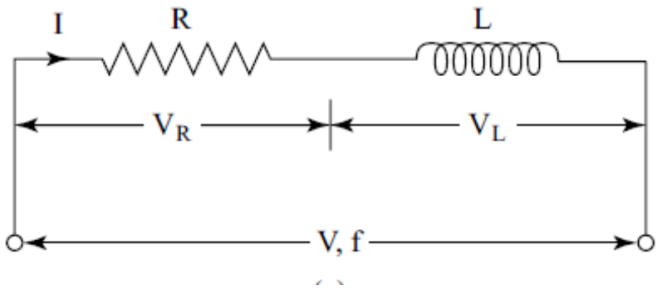
Module 2 – Lecture 2

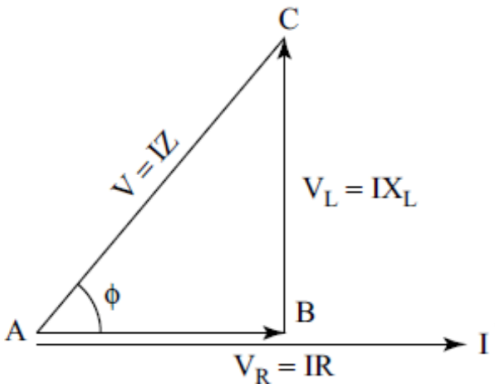
Dr. Arindam Dutta

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R—L Series Circuit





$V_R = IR$, $V_L = IX_L$ where $X_L = \omega L = 2\pi fL$

A voltage, V of frequency, f is applied across the circuit. The voltage drop across the resistance is V_R and across the inductor is V_L. Current flowing through the circuit is I. To get V, V_R and V_L are to be added vectorially as they are all not in phase, i.e., they are not along the same direction. To draw the current and voltage phasor we take the current I as the reference phasor since current I is common to V_R and V_L. V_R and I are in phase. V_L leads I by 90°. The angle between V and I is called the power factor angle φ. Power factor is cos φ. Considering the triangle ABC:

$$V^2 = V_R^2 + V_L^2$$

or,
$$V = \sqrt{V_R^2 + V_L^2} = \sqrt{(IR)^2 + (IX_L)^2}$$
$$= I\sqrt{R^2 + X_L^2}$$

or,
$$I = \frac{V}{\sqrt{R^2 + X_L^2}} = \frac{V}{Z} \text{ or, } V = IZ$$

where
$$Z = \sqrt{R^2 + X_L^{\ 2}}$$

V, V_R, V_L and I are RMS values

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R—L Series Circuit

Z is called the impedance of the total circuit. Triangle ABC is called the impedance triangle which is redrawn as shown below. From the impedance triangle:

$$Z = \sqrt{R^2 + X_L^2}$$

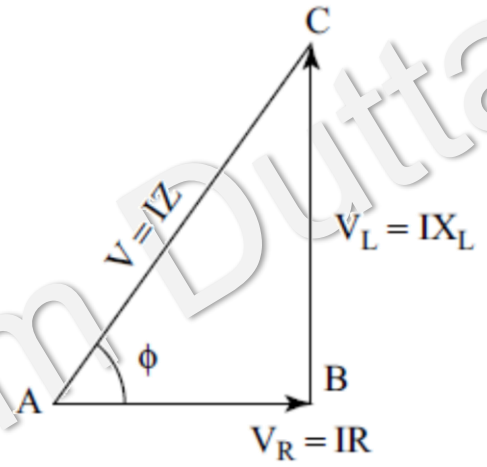
or, $$Z = R + jX_L$$

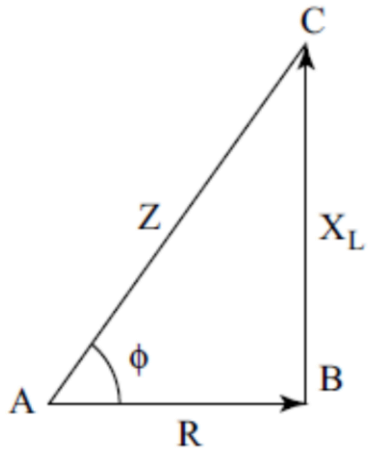
Where j indicates rotation by 90° in anti-clockwise direction.

$$\cos\phi = \frac{R}{Z}$$

Or, $$Z\cos\phi = R$$

and $$Z\sin\phi = X_L$$





The current, I has been kept aside which is common to all the sides. Impedance Z can be represented as the vector sum of R and X_L. since IX_L is leading I by 90° and R is in phase with I, we can write

$$Z = R + jX_L, \text{ and } \cos\phi = \frac{R}{Z}, \tan\phi = \frac{V_L}{V_R} = \frac{IX_L}{IR} = \frac{X_L}{R}$$

$$\phi = \tan^{-1}\frac{X_L}{R}, \cos\phi = \frac{R}{Z}$$

Power is, $$P = VI\cos\phi$$

or, $$\text{Power} = \text{Volt–Ampere} \times \text{Power factor}$$

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Apparent Power, Real Power, and Reactive Power

Apparent power (S): The product of the RMS value of voltage (V) and current (I). It is denoted by S.

Apparent power, S = VI = Voltage × Current VA or kVA

Real or true power or active power (P or W): It is the power which is actually dissipated in the circuit resistance (watt-full power).

Active power, P = Apparent power × power factor

$P = VI \cos\varphi$ Watts or kW

Reactive power (Q): It is the power developed in the inductive reactance of the circuit (watt-less power).

$Q = I^2 X_L = I^2 Z \sin\varphi = I (ZI) \sin\varphi$

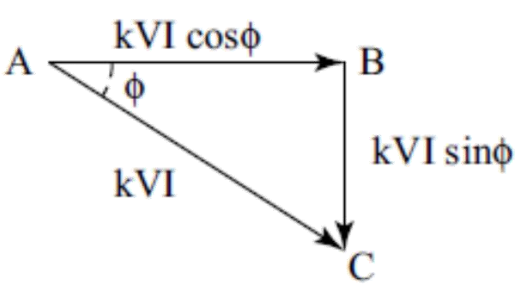
$Q = VI \sin\varphi$ VAR or kVAR

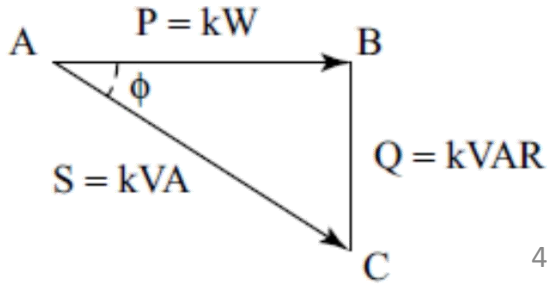
These three powers are shown in the form of a power triangle from where it can be seen that

$$S^2 = P^2 + Q^2$$

$$S = \sqrt{P^2 + Q^2}$$

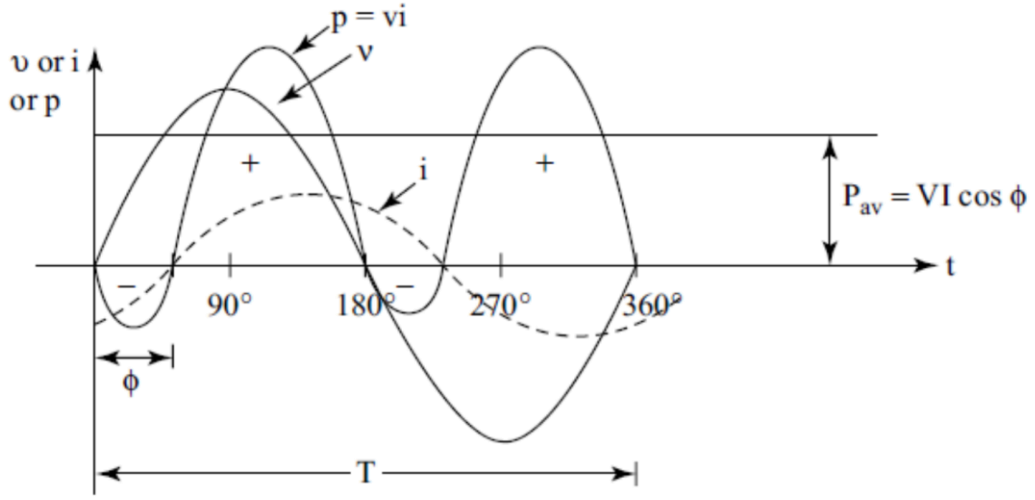
$$kVA = \sqrt{(kW)^2 + (kVAR)^2}$$

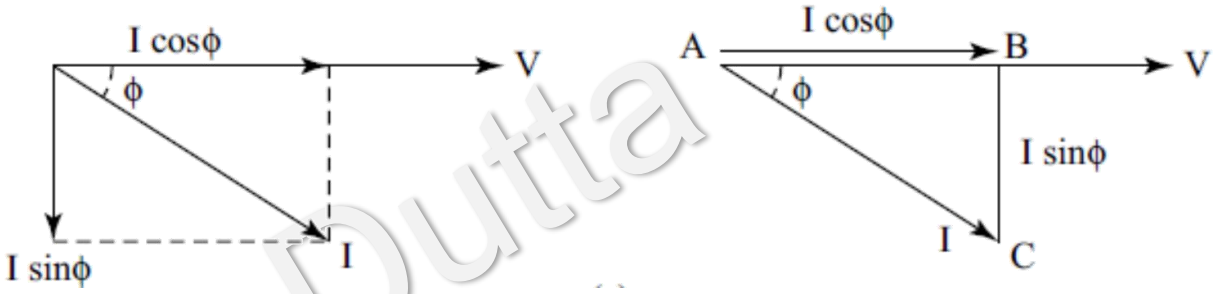




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Power in R—L Series Circuit





The expression for the voltage and current in series R–L circuit are $\mathbf{v = V_m \sin \omega t}$ and $\mathbf{i = I_m \sin (\omega t - \varphi)}$, as I lags V by an angle φ.

The average power over a complete cycle is calculated as

$$P_{av} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}} \cos\phi = V_{rms} \times I_{rms} \cos\phi = VI \cos\phi \, W$$

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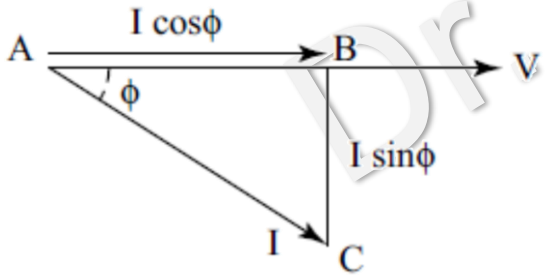
Power in R—L Series Circuit

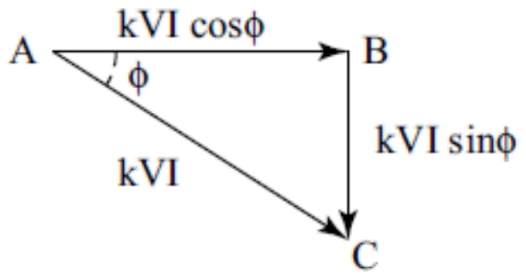
- **Power factor is the cosine of the phase angle (φ) between the voltage and current; cos φ. It is also expressed as the ratio, R/Z = resistance / impedance = cos φ.**

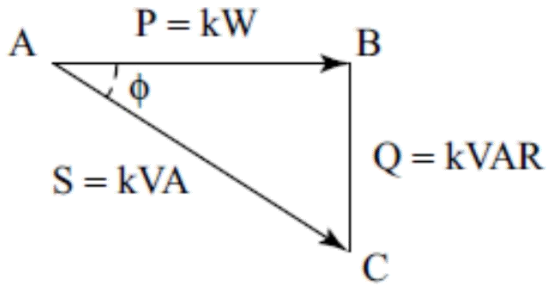
- In an R–L series circuit, I lags V by the power factor angle φ. The in-phase component of I is I cosφ and quadrature component is I sinφ Multiplying all the sides of the triangle ABC by kV (kilo-volt), we can draw the power triangle as in

$$kVA \cos\varphi = kW$$
$$kVA \sin\varphi = kVAR$$

- In the power triangle diagram, if φ is taken as zero, i.e., if the circuit is resistive, reactive power, Q becomes zero. If the circuit is having pure inductance or capacitance, φ = 90°, active power, P becomes zero. Reactive power will be present whenever there is inductance or capacitance in the circuit.

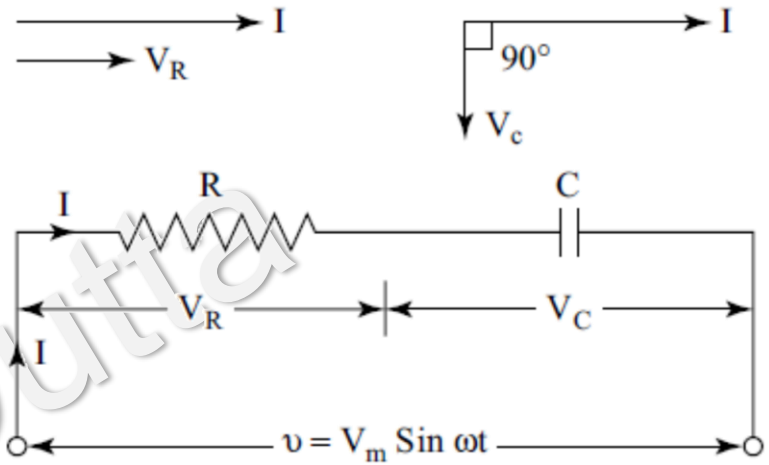






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R—C Series Circuit



- The voltage drop across the resistance is V_R and across the capacitor is V_C.
- $V_R = IR$, $V_C = IX_C$, where $X_C = 1/\omega C = 1/2\pi fC$
- I leads V by an angle φ or supply voltage V lags current I by an angle φ.

$$\tan\phi = \frac{V_C}{V_R} = \frac{IX_C}{IR} = \frac{X_C}{R}$$

$$\phi = \tan^{-1}\frac{X_C}{R}$$

Applied voltage,

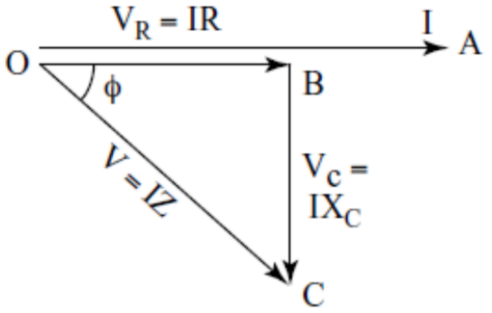
$$V = \sqrt{V_R^2 + V_C^2}$$

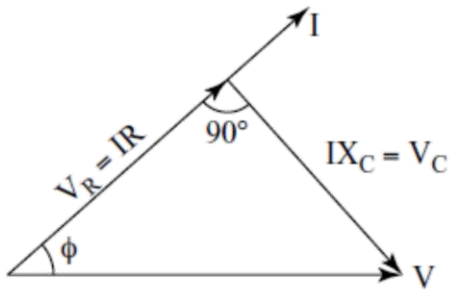
$$= \sqrt{(IR)^2 + (IX_C)^2}$$

$$= I\sqrt{R^2 + X_C^2}$$

$$V = IZ$$

where $Z = R - jX_c$ or $Z = \sqrt{R^2 + X_C^2}$ = impedance of the circuit





- **V, V_R, V_C and I are RMS values**

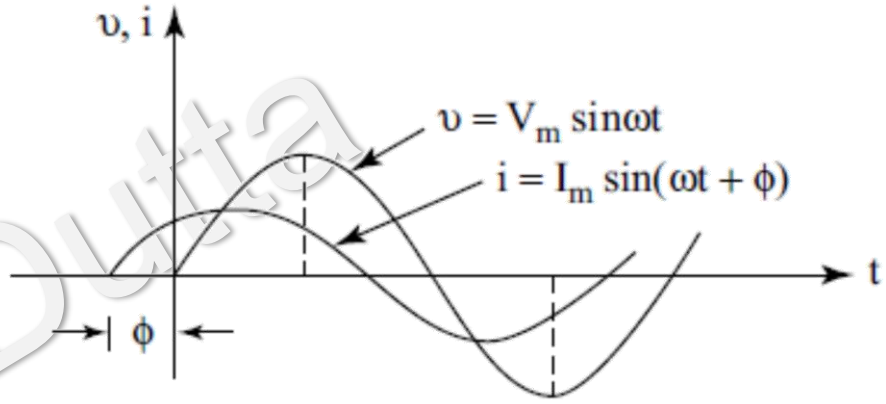
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Power in R—C Series Circuit

The expression for voltage and current is

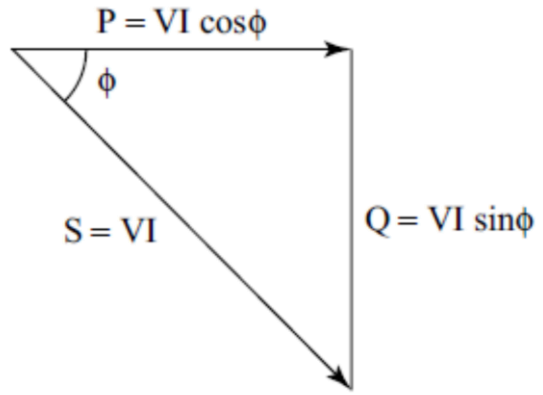
$$v = V_m \text{ Sin } \omega t$$

$$i = I_m \text{ Sin } (\omega t + \phi) \text{ as I leads V}$$



The average power over a complete cycle is calculated as

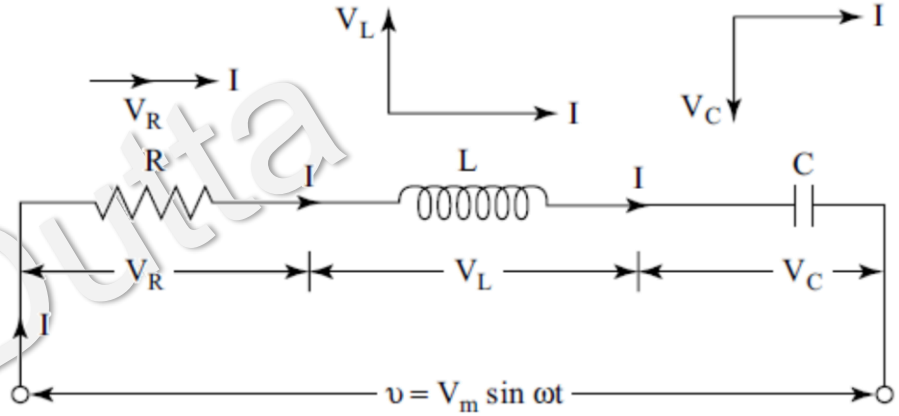
$$P_{av} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}} \cos\phi = V_{rms} \times I_{rms} \cos\phi = VI \cos\phi \text{ W}$$

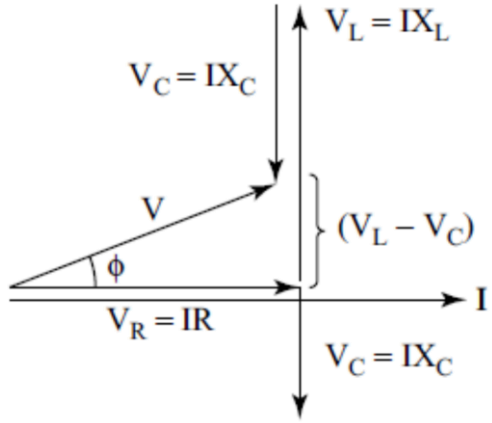


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R—L—C Series Circuit

- The voltage drop across resistance is V_R, across inductor is V_L, and across capacitor is V_C.
- $V_R = IR$, $V_L = IX_L$, where $X_L = \omega L = 2\pi fL$, $V_C = IX_C$, where $X_C = 1/\omega C = 1/2\pi fC$
- **V, V_R, V_L, V_C and I are RMS values**
- The phasor diagram depends on the magnitude of V_L and V_C, which depends on X_L and X_C. So, three different cases can be considered:
  - (a) When $X_L > X_C$
  - (b) When $X_L < X_C$
  - (c) When $X_L = X_C$





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R—L—C Series Circuit ($X_L > X_C$)

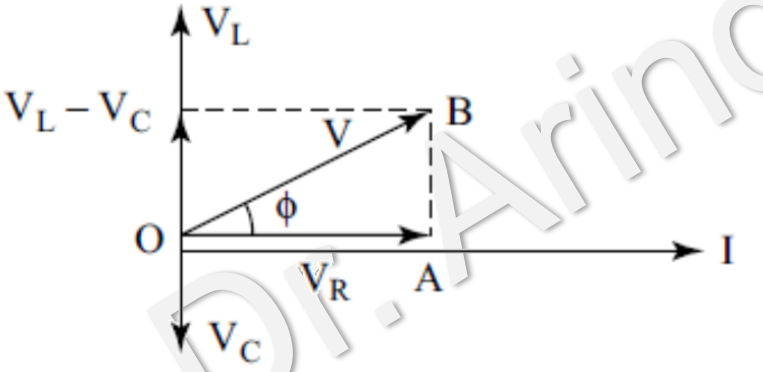
The circuit will effectively be **inductive** in nature and I will lag V. IX_L, i.e., V_L is greater than IX_C, i.e., V_C. So, the resultant of V_L and V_C will be $V_L - V_C$ so that V is the phasor sum of V_R and $(V_L - V_C)$.

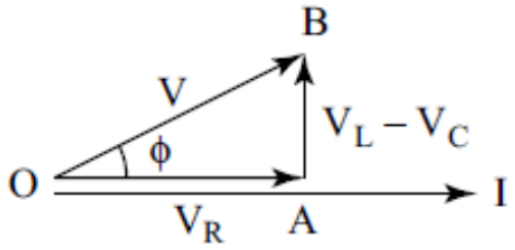
$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$= \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$

$$= I\sqrt{R^2 + (X_L - X_C)^2}$$

$$V = IZ$$





$V_R = V\cos\phi$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\tan\phi = \frac{(X_L - X_C)}{R}, \quad \phi = \tan^{-1}\frac{(X_L - X_C)}{R}$$

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R—L—C Series Circuit ($X_L < X_C$)

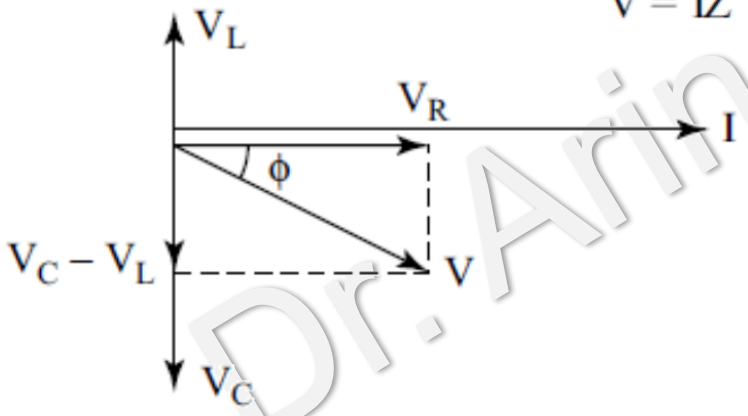
The circuit will effectively be **capacitive** in nature and I will lead V. IX_C, i.e., V_C is greater than IX_L, i.e., V_L. So, the resultant of V_L and V_C will be $V_C - V_L$ so that V is the phasor sum of V_R and $(V_C - V_L)$.

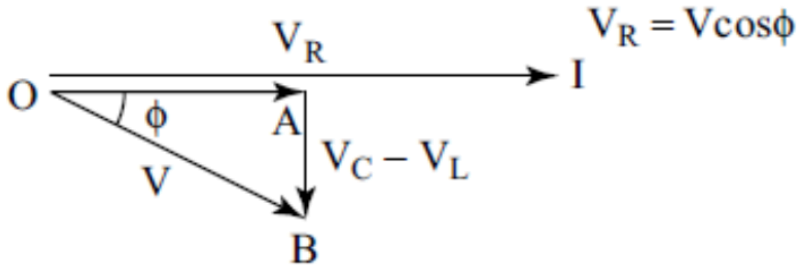
$$V = \sqrt{V_R^2 + (V_C - V_L)^2}$$

$$= \sqrt{(IR)^2 + (IX_C - IX_L)^2}$$

$$= I\sqrt{R^2 + (X_C - X_L)^2}$$

$$V = IZ$$





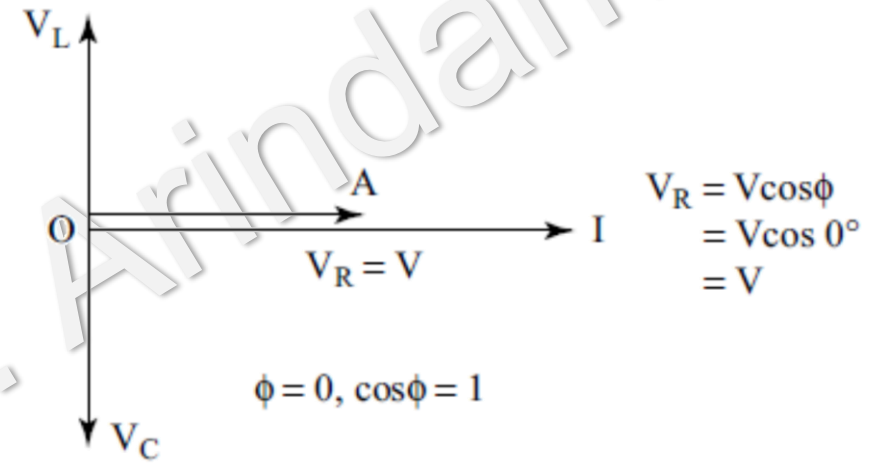
$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

$$\phi = \tan^{-1}\left(\frac{X_C - X_L}{R}\right)$$

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R—L—C Series Circuit ($X_L = X_C$)

- When $X_L = X_C$, V_L and V_C will cancel each other and their resultant will be zero.
- The circuit will effectively be **resistive** in nature and I will be in phase with V.
- The impedance of the circuit will be minimum, i.e., equal to R.



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Power in R—L—C Series Circuit

The average power consumed by the circuit is

P_{av} = Average power consumed by R + Average power consumed by L + Average power consumed by C.

But pure L and C never consume any power.

Therefore, P_{av} = Power taken by R = $I^2R = IV_R$

But $V_R = V\cos\phi$ in all the cases.

Therefore, $P = VI\cos\phi$ W.

Thus, for any condition, that is when $X_L > X_C$ or $X_L < X_C$ or $X_L = X_C$, power can be expressed as P = Voltage × Component of I in phase with V = $VI\cos\phi$

Note that when $X_L = X_C$, the component of I in phase with V is I only because $I\cos\phi = I$ (as $\cos\phi = 1$).

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Resonance in AC circuits

- In ac circuits resonance occurs when two independent energy storing devices are capable of interchanging energy from one another

- When frequency is variable, the inductive and capacitive reactance of the circuit elements

$$\mathbf{X_L = 2\pi fL} \text{ and } \mathbf{X_C = 1/2\pi fC}$$

will change

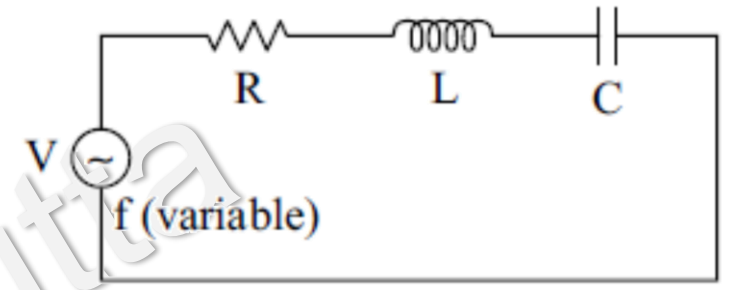
- The current in the circuit will depend upon the values of X_L and X_C and that of R

- A condition may occur at a particular frequency that the impedance offered to the flow of current is maximum or minimum

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Resonance in AC series circuits

The supply voltage is constant but its frequency is variable



The impedance of the circuit, $Z = R + j\omega L - j\frac{1}{\omega C}$

The current flowing through the circuit is

$$I = \frac{V}{R + j\left(\omega L - \frac{1}{\omega C}\right)} = \frac{V}{R + j(X_L - X_C)}$$

$$= \frac{V}{R + j\left(2\pi fL - \frac{1}{2\pi fC}\right)}$$

As the frequency is changing, both X_L and X_C will change. Inductive reactance X_L will increase as the frequency, f is increasing while the capacitive reactance, X_C will decrease with increasing frequency. The value of R is independent of frequency.

At some frequency f_0, the magnitude of X_L will be equal to X_C. Since X_L and X_C are vectorially jX_L and $-jX_C$, the two reactances will cancel each other when frequency is f_0. At f_0 the impedance of the circuit is equal to R which is the minimum value of Z. So, at $f = f_0$ maximum current ($I = V/R$) will flow through the circuit.

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Resonance in AC series circuits

This condition of the circuit when X_L equals X_C, Z = R, and current is maximum is called the resonant condition and the frequency, f_0 at which resonance occurs is called the resonant frequency. At resonance, since X_L equals X_C,

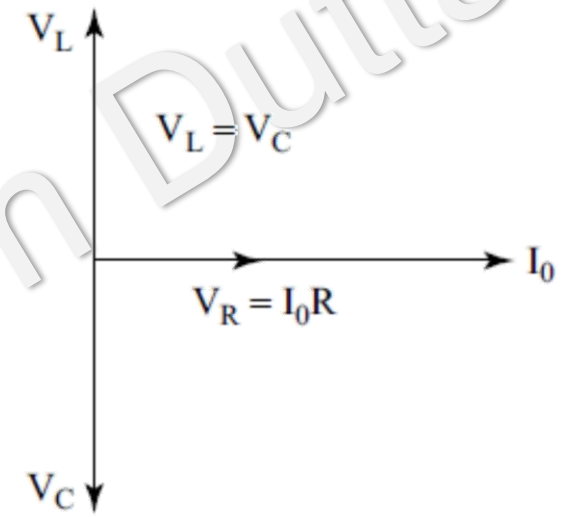
$$2\pi f_0 L = \frac{1}{2\pi f_0 C}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\omega_0^2 = \frac{1}{LC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$



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At resonance, frequency is f_0, current $I_0 = V/R$, and power factor ($\cos\varphi$) is unity.

As $V_L = V_C$ and they are in phase opposition, the net voltage across L and C in series is equal to zero. Thus, the supply voltage will be equal to V_R.

Since at resonance, current is maximum and is very high, power dissipation I_0^2R is maximum and the rate of energy storage in the inductor and the capacitor is maximum and they are equal.

Problem

Q: An R–L–C series circuit has R = 10 Ω, L = 0.1 H, and C = 8 μF. Calculate the resonant frequency.

For resonance $X_L = X_C$

$$2\pi f_0 L = \frac{1}{2\pi f_0 C}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$f_0 = \frac{1}{2\pi\sqrt{0.1\times 8\times 10^{-6}}} = 178\,\text{Hz}$$

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Problem

Q: A resistor, a variable iron-core inductor, and a capacitor are connected across a 230 V, 50 Hz supply. By varying the position of the iron core inside the inductor coil, its inductance is changed. Maximum current of 1.5 A was obtained in the circuit by changing the inductance of the coil. At that time the voltage across the capacitor was measured as 600 V. Calculate the values of circuit parameters.

A: Maximum current flows at resonance when $X_L = X_C$ and $Z = R$

Maximum current, $$I_0 = I_m = \frac{V}{Z} = \frac{V}{R} [\because Z = R]$$

Therefore, $$R = \frac{V}{I_m} = \frac{230}{1.5} = 153.3\ \Omega$$

At resonance, $$V_L = V_C = 600\ V$$

$$V_L = I_m X_L = 600\ V$$

$$X_L = \frac{600}{1.5} = 400\ \Omega$$

or, $$2\pi fL = 400$$

$$L = \frac{400}{6.28 \times 50} = 1.27\ H$$

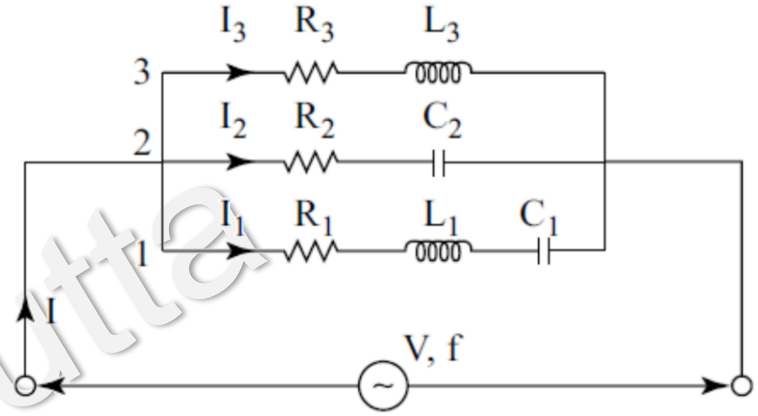
$$X_C = X_L = 400$$

$$\frac{1}{2\pi fC} = 400$$

$$C = \frac{1}{314 \times 400} = 7.96 \times 10^{-6}\ F$$

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AC Parallel Circuits



Branch 1

Impedance of branch 1, $= \sqrt{(R_1)^2 + (X_{L1} - X_{C1})^2} = Z_1$

Current $I_1 = V/Z_1$

Phase difference of this current with respect to the applied voltage is given by $\phi_1 = \tan^{-1} \dfrac{(X_{L1} - X_{C1})}{R_1}$

This current will lag the applied voltage by an angle ϕ_1, if $X_{L1} > X_{C1}$.

In case $X_{C1} > X_{L1}$, then I_1, will lead V.

Branch 2

Capacitive branch (I_2 leads V)

Impedance of branch 2, $Z_2 = \sqrt{(R_2)^2 + (X_{C2})^2}$

Current $I_2 = V/Z_2$

The branch current I_2 leads the applied voltage V, by an angle ϕ_2, given by

$$\phi_2 = \tan^{-1} \frac{(X_{C2})}{R_2}$$

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AC Parallel Circuits

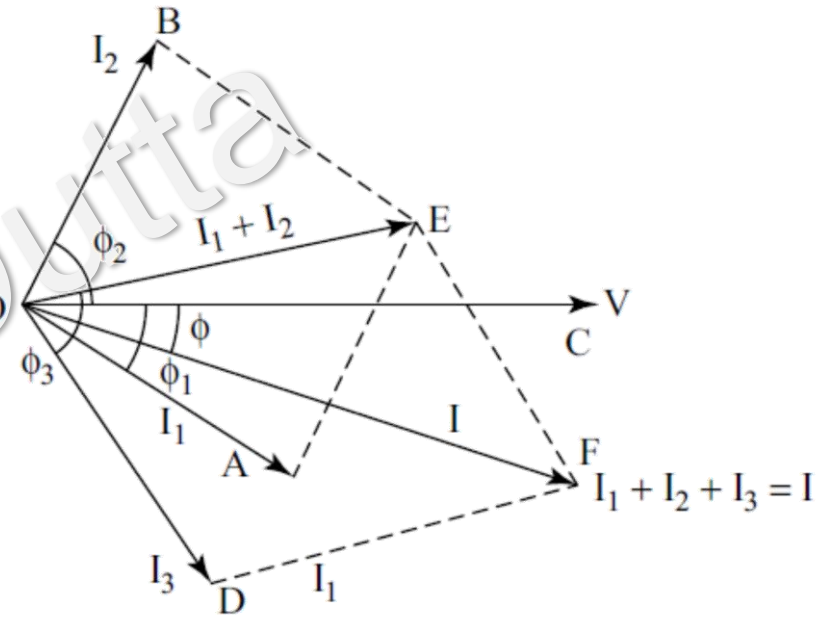
Branch 3

Inductive branch 3, $Z_3 = \sqrt{(R_3)^2 + (X_{L3})^2}$

Current $I_3 = V/Z_3$

This current will lag the applied voltage by an angle ϕ_3,

$$\phi_3 = \tan^{-1}\frac{X_{L3}}{R_3}$$



The equivalent impedance of the circuit can be found out by

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}$$

Therefore, the line current, $I = V / Z$

Alternatively, the line current can be calculated by, $I = I_1 + I_2 + I_3$ (vector addition)

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AC Parallel Circuits

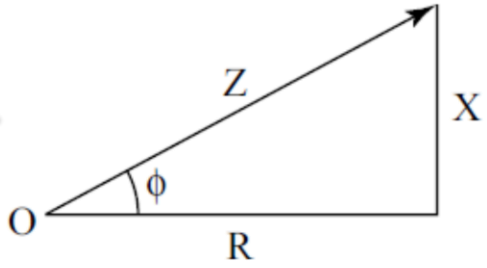
Admittance: It is the reciprocal of impedance, denoted by Y and is measured in unit mho or Siemens.

Components of admittance:

If the circuit contains R and L, $Z = R + j X_L$;

If the circuit contains R and C, $Z = R - jX_C$.

Considering X_L and X_C as X we can write $Z = R \pm jX$.



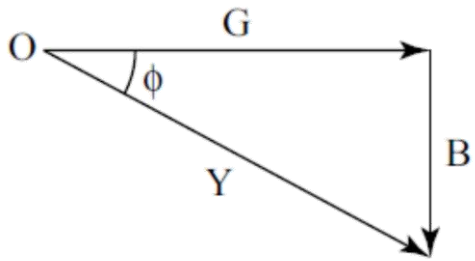
Impedance Triangle

Consider an impedance as given by

$$Z = R \pm jX$$

Positive sign is for an inductive circuit and negative sign is for a capacitive circuit.

Admittance $$Y = \frac{1}{Z} = \frac{1}{R \pm jX}$$



Admittance Triangle

Rationalizing the above expression,

$$Y = \frac{R \mp jX}{(R \pm jX)(R \mp jX)} = \frac{R \mp jX}{R^2 + X^2} = \frac{R}{R^2 + X^2} \mp j\frac{X}{R^2 + X^2} = \frac{R}{Z^2} \mp j\frac{X}{Z^2} = G \mp jB$$

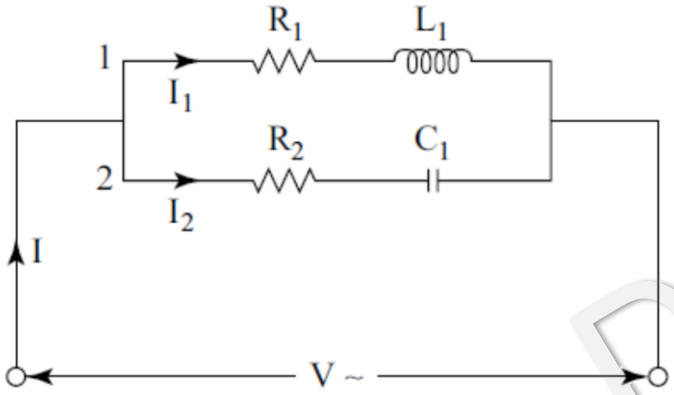
$G = R/Z^2$ mho = conductance; $B = X/Z^2$ mho = susceptance

B is negative if the circuit is inductive and B is positive if the circuit is capacitive.

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AC Parallel Circuits

Admittance Method



Consider a parallel circuit consisting of two branches 1 and 2. Branch 1 has R_1 and L_1 series while Branch 2 has R_2 and C_1 series, respectively. The voltage applied to the circuit is V Volts

Total conductance is found by adding the conductances of two branches. Similarly, the total susceptance is found by algebraically adding the individual susceptance of different branches.

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2} \qquad I_1 = \frac{V}{Z_1} = VY_1$$

$$Y = Y_1 + Y_2 \qquad I_2 = \frac{V}{Z_2} = VY_2$$

Total conductance	$G = G_1 + G_2$
Total susceptance	$B = (-B_1) + B_2$
Total current	$I = VY$
Power factor	$\cos\phi = G/Y$

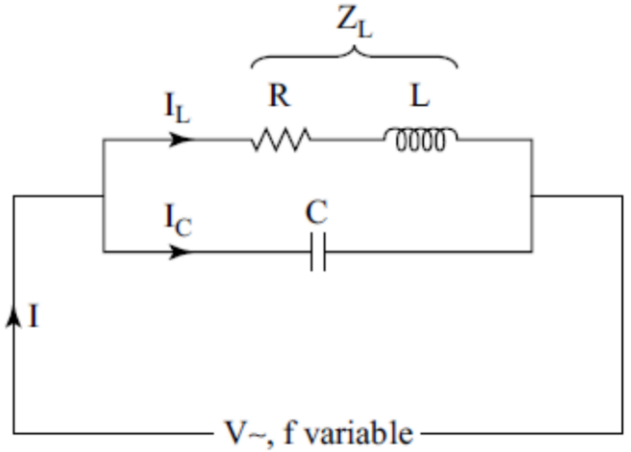
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Resonance in AC Parallel Circuits

Let us consider an inductive coil and a capacitor in parallel connected across a constant voltage variable frequency supply source. Practically, both the capacitor and the inductor will have some losses represented by a small resistance in series. Here we assume the capacitor as a loss-less one while the inductor coil has some resistance which has been shown separately.

The resonance frequency of a parallel L and C circuit is $f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$

If we consider the value of R as negligible, then the resonance frequency is $f_0 = \frac{1}{2\pi\sqrt{LC}}$



The impedance at resonance is given by $Z_0 = \frac{L}{CR}$

The line current is given by $I = \frac{V}{Z_0} = \frac{V}{L/CR} = \frac{VCR}{L}$

The line current I is the minimum current occurring at resonance. The circuit behaves like a resistive circuit only since the reactive component currents cancel each other. Z_0 is therefore resistive only. Since current is minimum, Z_0 is maximum under the resonant condition.

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Problem

Q: An inductive coil has a resistance of 10 Ω and inductance of 100 mH. This coil is connected in parallel with a capacitor of 20 μF. A variable frequency power at 100 V is applied across this parallel circuit. Calculate the frequency at which the circuit will resonate. Also calculate the dynamic impedance of the circuit and resonant current.

Resonant frequency, $$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

Substituting the given values,

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{100\times10^{-3}\times20\times10^{-6}} - \frac{(10)^2}{(100\times10^{-3})^2}}$$

$$= 112.48 \text{ Hz}$$

Dynamic impedance, $$Z_0 = \frac{L}{CR} = \frac{100\times10^{-3}}{20\times10^{-6}\times10} = 500\ \Omega$$

Current at resonance, $$I_0 = \frac{V}{Z_0} = \frac{100}{500} = 0.2 \text{ A}$$

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AC Series-Parallel Circuits

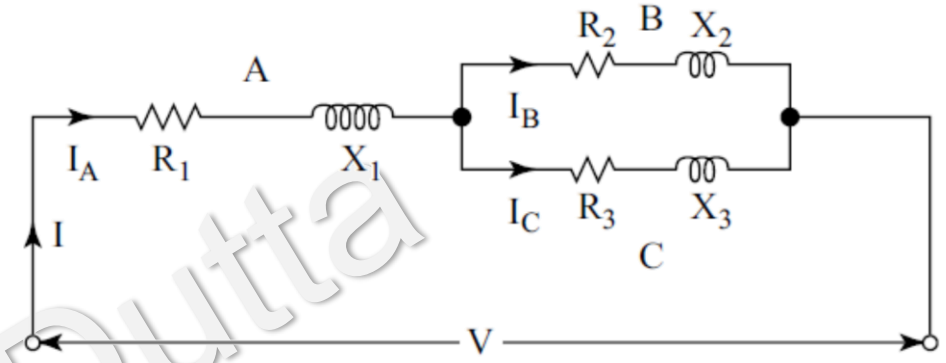
Impedance of branch A, $Z_A = R_1 + jX_1$

Impedance of branch B, $Z_B = R_2 + jX_2$

Impedance of branch C, $Z_C = R_3 + jX_3$

Total impedance of the circuit Z is

$$Z = Z_A + \frac{Z_B Z_C}{Z_B + Z_C}$$



Total current, $I = \frac{V}{Z} = I_A$

Current, $I_B = I \frac{Z_C}{Z_B + Z_C}$ (applying current divider rule)

Current, $I_C = I \frac{Z_B}{Z_B + Z_C}$

By applying the admittance method we can also solve the problem as

$$Y_A = \frac{1}{Z_A}; Y_B = \frac{1}{Z_B}; Y_C = \frac{1}{Z_C}$$

Total admittance of the parallel branches B and C $Y_{BC} = Y_B + Y_C$

Impedance $Z_{BC} = \frac{1}{Y_{BC}}$

Total Impedance $Z = Z_A + Z_{BC}$

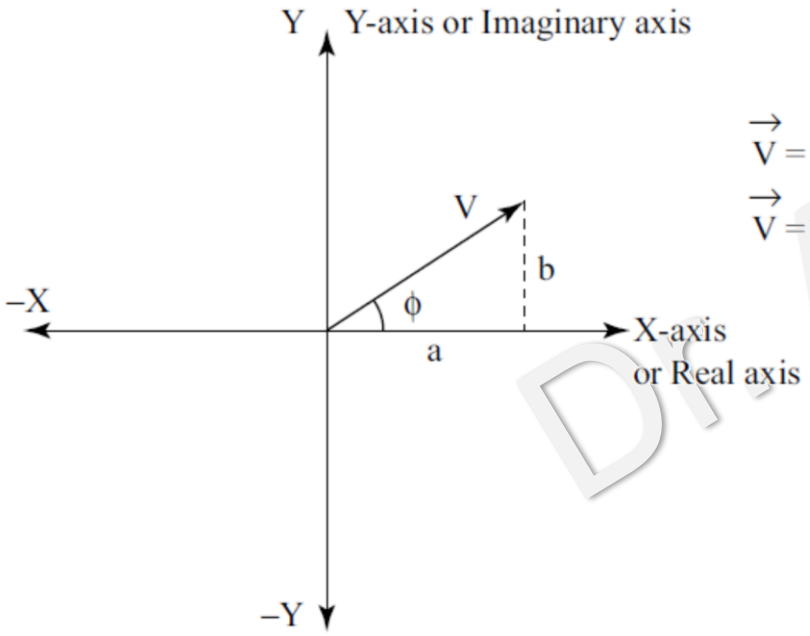
Total current $I = \frac{V}{Z} = \frac{V}{Z_A + Z_{BC}}$

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Vector Algebra

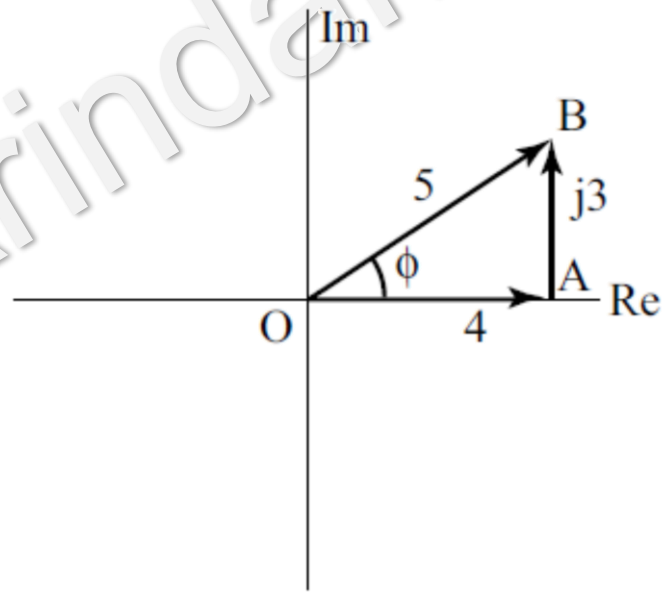
The voltage, V can be represented as $V\angle\phi$. This is called the polar form of representation. Voltage V can also be represented as $V = a + jb = V\cos\phi + jV\sin\phi$. This is called the rectangular form of representation using a j operator.

As shown in the figure, phasor OB can be represented as $5\angle\phi$ in the polar form. In the rectangular form OB is represented as $4 + j3$



$\vec{V} = V\angle\phi$

$\vec{V} = a + j\,b$



$$\overrightarrow{OB} = \left|\sqrt{OA^2 + AB^2}\right|\angle\tan^{-1}\frac{AB}{OA}$$

$$= \sqrt{4^2 + 3^2}\angle\tan^{-1}\frac{3}{4}$$

$$= 5\angle 37^\circ = 5\cos 37^\circ + j\,5\sin 37^\circ$$

$$= 5\times 0.8 + j5\times 0.6 = 4 + j3$$

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Vector Algebra

Addition and subtraction of phasor quantities

Let $V_1 = a_1 + jb_1$ and $V_2 = a_2 + jb_2$

Addition: $V = V_1 + V_2$

$= (a_1 + jb_1) + (a_2 + jb_2)$

$= (a_1 + a_2) + j(b_1 + b_2)$

Magnitude of the resultant vector $V = \sqrt{(a_1 + a_2)^2 + (b_1 + b_2)^2}$

Phase angle, $\theta = \tan^{-1} = \dfrac{(b_1 + b_2)}{(a_1 + a_2)}$

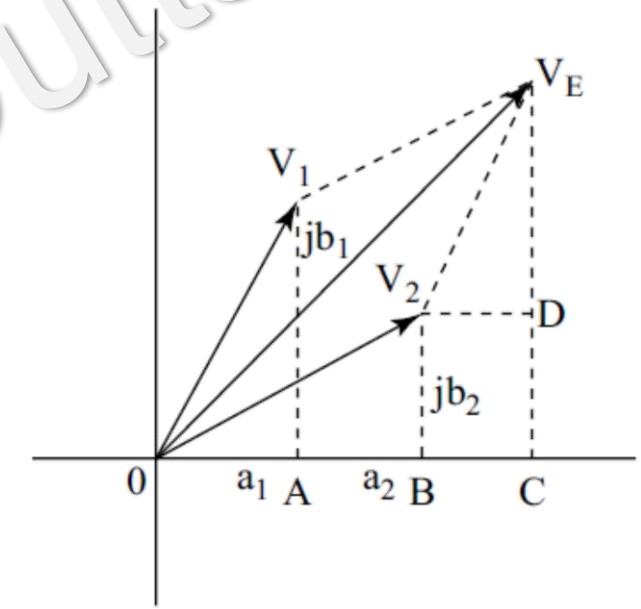
Subtraction: $V = V_1 - V_2$

$= (a_1 + jb_1) - (a_2 - jb_2)$

$= (a_1 - a_2) + j(b_1 - b_2)$

Magnitude of the resultant vector $V = \sqrt{(a_1 - a_2)^2 + (b_1 - b_2)^2}$

Phase angle, $\theta = \tan^{-1} = \dfrac{(b_1 - b_2)}{(a_1 - a_2)}$



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Vector Algebra

Multiplication and division of phasor quantities

Let $V_1 = a_1 + jb_1 = V_1\angle\theta_1$; where $\theta_1 = \tan^{-1}\left(\frac{b_1}{a_1}\right)$

$V_2 = a_2 + jb_2 = V_2\angle\theta_2$; where $\theta_2 = \tan^{-1}\left(\frac{b_2}{a_2}\right)$

Multiplication

$V_1 = V_1 \times V_2 = V_1\angle\theta_1 \times V_2\angle\theta_2$ in the polar form

$= V_1 V_2 \angle(\theta_1 + \theta_2)$, angles are added algebraically

Division

$$\frac{V_1}{V_2} = \frac{a_1 + jb_1}{a_2 + jb_2} = \frac{V_1\angle\theta_1}{V_2\angle\theta_2} = \frac{V_1}{V_2}\underline{\lfloor\theta_1 - \theta_2}$$

angles are subtracted algebraically

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Problem

Q: A coil having a resistance of 5 Ω and inductance of 30 mH in series are connected across a 230 V, 50 Hz supply. Calculate current, power factor, and power consumed.

$$R = 5\ \Omega$$

$$L = 30 \times 10^{-3}\ H$$

Inductive reactance, $X_L = \omega L = 2\pi f L$

$$X_L = 2 \times 3.14 \times 50 \times 30 \times 10^{-3}\ \Omega$$

$$= 9.42\ \Omega$$

Impedance, $Z = R + jX_L = 5 + j9.42$

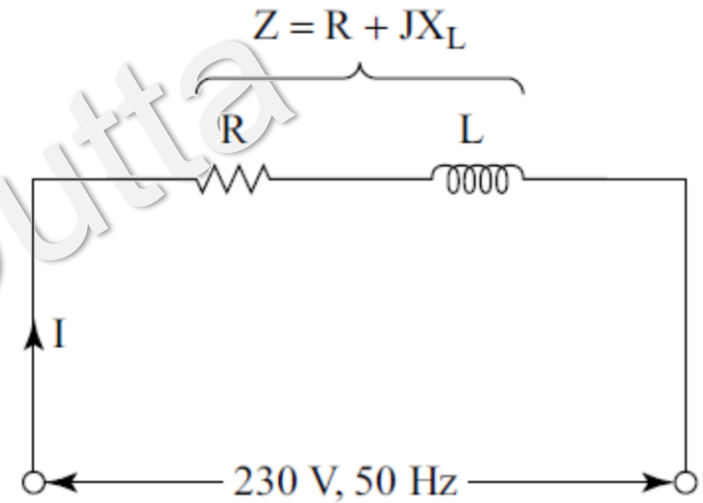
$$= \sqrt{5^2 + (9.42)^2} \angle \tan^{-1} \frac{9.42}{5}$$

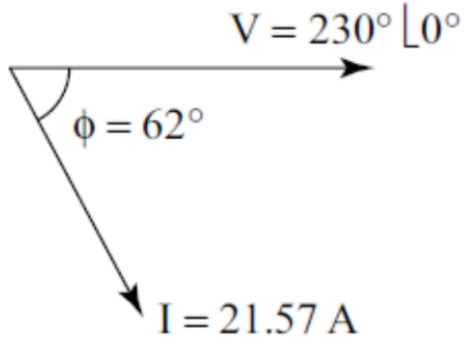
$$= 10.66 \angle \tan^{-1} 1.884$$

$$= 10.66 \angle 62^\circ\ \Omega$$

Current, $I = \frac{V}{Z} = \frac{230 \angle 0}{10.66 \angle 62^\circ} = 21.57 \angle -62^\circ A$

Magnitude of $I = 21.57\ A$





Current I is lagging the voltage, V by 62°. Power factor = cos {(0° – (–62°)} = cos (62°) = 0.47 lagging.

Power consumed = VI cosφ = 230 × 21.57 × 0.47 W = 2331.7 W

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Problem

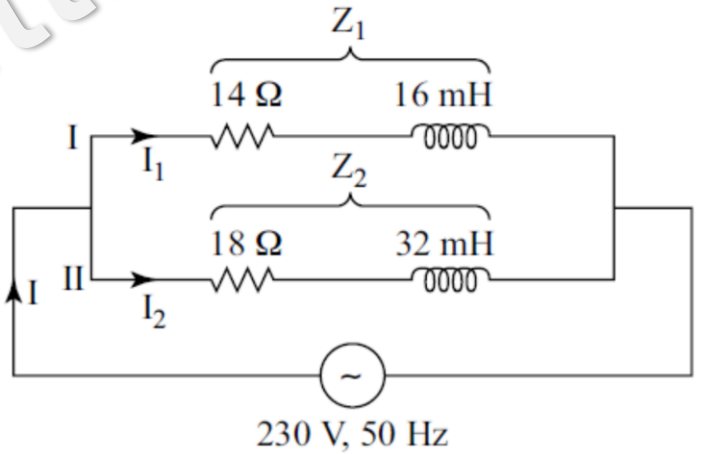
Q: Two impedances Z_1 and Z_2 are connected in parallel across a 230 V, 50 Hz supply. The impedance, Z_1 consists of a resistance of 14 Ω and an inductance of 16 mH. The impedance, Z_2 consists of a resistance of 18 Ω and an inductance of 32 mH. Calculate the branch currents, line current, and total power factor. Draw the phasor diagram showing the voltage and currents.

Let $R_1 = 14\ \Omega,\ X_L = \omega L_1 = 2\pi f L_1 = 2 \times 3.14 \times 50 \times 16 \times 10^{-3} = 5\ \Omega$

$$|Z_1| = \sqrt{R_1^2 + X_{L_1}^2} = \sqrt{14^2 + 5^2} = 14.9\ \Omega$$

$R_2 = 18\ \Omega,\ X_{L2} = \omega L_2 = 2\pi f L_2 = 2 \times 3.14 \times 50 \times 32 \times 10^{-3} = 10\ \Omega$

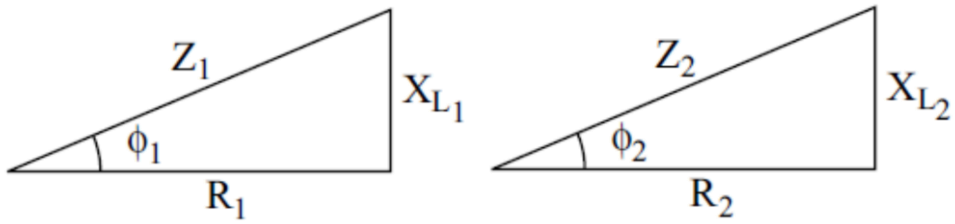
$$|Z_2| = \sqrt{R_2^2 + X_{L_2}^2} = \sqrt{18^2 + 10^2} = 20.6\ \Omega$$



The phase angles of Z_1 and Z_2 are calculated from the impedance triangles as

$$\phi_1 = \tan^{-1}\frac{X_{L1}}{R_1} = \tan^{-1}\frac{5}{14} = 19.6°$$

$$\phi_2 = \tan^{-1}\frac{X_{L2}}{R_2} = \tan^{-1}\frac{10}{18} = 29°$$



Thus, $Z_1 = 14.9\angle 19.6°$ and $Z_2 = 20.6\angle 29°$

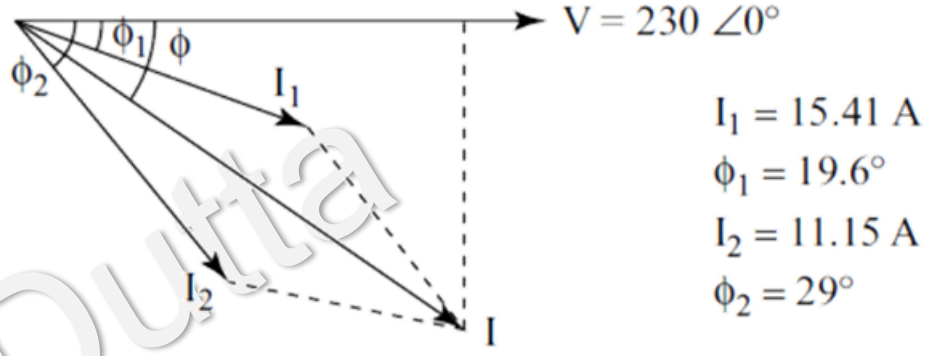
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Problem

Admittance of branch I is Y_1 and admittance of branch II is Y_2

$$Y_1 = \frac{1}{Z_1} = \frac{1}{14.9\angle 19.6^\circ} = 0.067\angle -19.6^\circ$$

$$Y_2 = \frac{1}{Z_2} = \frac{1}{20.6\angle 29^\circ} = 0.0485\angle -29^\circ$$



$I_1 = 15.41$ A
$\phi_1 = 19.6^\circ$
$I_2 = 11.15$ A
$\phi_2 = 29^\circ$

Taking voltage, V as the reference axis,

$$I_1 = \frac{V}{Z_1} = VY_1 = 230\angle 0 \times 0.067\angle -19.6^\circ = 15.41\angle -19.6^\circ \text{ A}$$

$$I_2 = \frac{V}{Z_2} = VY_2 = 230\angle 0^\circ \times 0.0485\angle -29^\circ = 11.15\angle -29^\circ \text{ A}$$

Line current, $I = I_1 + I_2 = 15.41\angle -19.6^\circ + 11.15\angle -29^\circ$

$= (14.52 - j5.17) + (9.75 - j5.41)$

$= (24.27 - j10.58)$ A

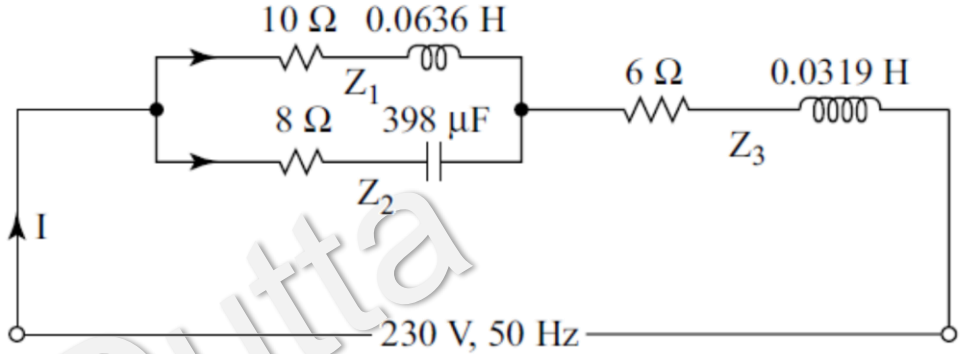
$= 26.48\angle -23.55^\circ$ A

Power factor = cos {0° – (–23.55°)} = cos (23.55°) = 0.917 lagging.

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Problem

Q: Determine the total current drawn from the supply by the series–parallel circuit. Also calculate the power factor of the circuit.



$$\omega = 2\pi f = 2 \times 3.14 \times 50 = 314$$

$$Z_1 = 10 + j314 \times 0.0638 = 10 + j20 = 22.36 \angle 64°$$

$$Z_2 = 8 - jX_C$$

$$X_C = \frac{1}{\omega C} = \frac{10^6}{314 \times 398} = 8\,\Omega$$

$$Z_2 = 8 - j8 = 11.3 \angle -45°$$

$$Z_3 = 6 + j314 \times 0.0319 = 6 + j10 = 11.66 \angle 59°$$

$$Z = \frac{Z_1 Z_2}{Z_1 + Z_2} + Z_3$$

$$= \frac{22.36 \angle 64° \times 11.3 \angle -45°}{10 + j20 + 8 - j8} + 11.66 \angle 59°$$

$$= \frac{252.7 \angle 19°}{18 + j12} + 11.66 \angle 59°$$

$$= \frac{252.7 \angle 19°}{21.63 \angle 34°} + 11.66 \angle 59°$$

$$= 11.68 \angle -15° + 11.66 \angle 59°$$

$$= 11.21 - j3 + 6 + j10$$

$$= 17.21 + j7$$

$$= 18.58 \angle 22°\,\Omega$$

$$I = \frac{V}{Z} = \frac{230 \angle 0°}{18.58 \angle 22°} = 12.37 \angle -22°\,A$$

Total current, $I = 12.37$ A

Power factor, $\cos\phi = \cos 22° = 0.92$ lagging

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EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 2 – Lecture 3

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Responses in First Order Systems

Transient response or natural response refers to a change from an equilibrium or a steady state. It is a temporary behaviour of a system as it adjusts to a new input or disturbance.

A **steady-state response or forced response** is the behavior of a circuit after a long time when steady conditions have been reached after an external excitation.

In the previously discussed method for AC circuit analysis, all voltages and currents are represented as phasors and all circuit components (R, C, and L) are represented by their impedances, so that we can solve the corresponding algebraic equations to get the steady-state responses of the circuit to an AC voltage or current input.

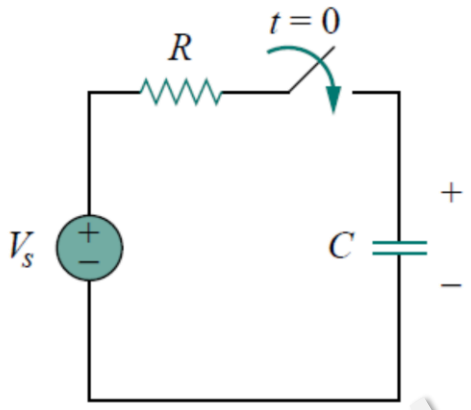
However, if we are also interested in the transient response of the circuit to an input which is turned on at time moment $t = 0$, we need to solve the corresponding differential equations to find its complete solution as the sum of the natural response and its forced response.

The forced response is what the circuit does with the sources turned on, but with the initial conditions (initial currents on inductors and initial voltage on capacitors for example) set to zero. The natural response is what the circuit does including the initial conditions, but with the input suppressed, that is, with all external forces set to zero. In circuits, this would be the response of the circuit with some initial condition and with all the independent voltages set to zero volts (short circuit) and current sources set to zero amps (open circuit). **The total or complete response is the sum of the forced response plus the natural response.**

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Responses in First Order Systems

The RC circuit is composed of a resistor R and capacitor C in series with an external voltage input $v(t)$, which is turned on at $t = 0$, either by a switch or a step voltage $v(t)$. We also assume the initial condition that the voltage across C is $v_C(0) = V_0$ at $t = 0$. Any of the variables $v_R(t)$, $v_C(t)$, and $i(t)$ can be considered as the circuit's response to this input.



At $t = 0$, energy stored by capacitor $= \frac{1}{2} C V_0^2$

- The RC circuit can be described by the KVL:

$$v_R(t) + v_C(t) = R\, i(t) + v_C(t) = RC \frac{d}{dt} v_c(t) + v_c(t) = \tau \frac{d}{dt} v_C(t) + v_C(t) = v(t)$$

where $\tau = RC$ is the time constant of the system with the dimension of time:

$$[RC] = \frac{[V]}{[I]} \frac{[Q]}{[V]} = \frac{[Q]}{[I]} = [T]$$

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Responses in First Order Systems

When the external input is zero $v_s(t) = 0$, the DE is homogeneous (zero on the right-hand side):

$$\tau \frac{d}{dt} v_C(t) + v_C(t) = 0, \qquad v_C(t)|_{t=0} = v_C(0) = V_0$$

Solving the DE gives the **natural/transient response** of the RC circuit as $\boxed{v_C(t) = V_0 e^{-t/\tau}}$

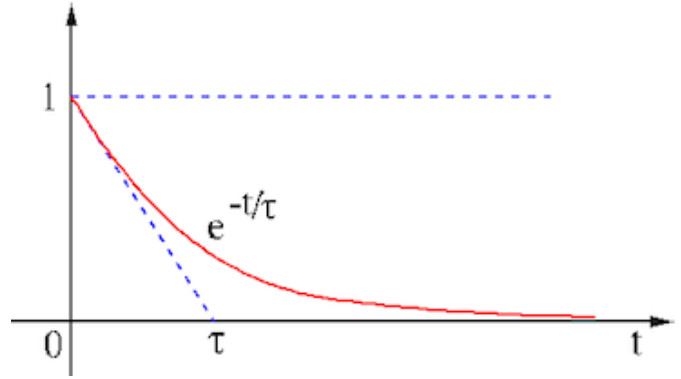
The current through R and C is

$$i(t) = C \frac{d}{dt} v_C(t) = C \frac{d}{dt} \left(V_0 e^{-t/\tau} \right) = -V_0 \frac{C}{\tau} e^{-t/\tau} = -\frac{V_0}{R} e^{-t/\tau}$$

The voltage across R is

$$v_R(t) = i(t)R = -V_0 e^{-t/\tau} = -v_C(t)$$

This result can be verified by KVL: $v_C(t) + v_R(t) = v_s(t) = 0$.



The time constant τ can be identified on the time plot of the general first order response $v(t) = V_0 e^{-t/\tau} = e^{-t/\tau}$ with $V_0 = 1$.

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Responses in First Order Systems

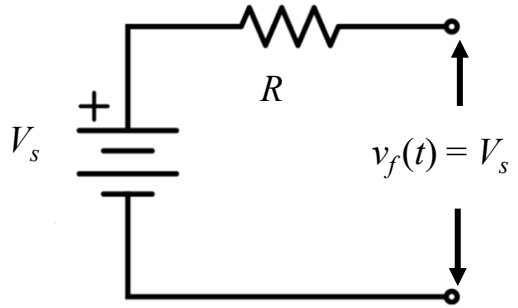
Complete Solution

Total Response = Natural Response + Forced Response

We want to solve the DE for $v_C(t)$ as the circuit's complete response to the input $v(t) \neq 0$ as well as the initial condition $v_C(0) = V_0 \neq 0$, after a switch is closed at time moment $t = 0$. We also want to find the voltage $v_R(t)$ across R as a function of time for $t > 0$.

We consider a constant (DC) input $v(t) = V_s$ (step input) being applied to the circuit at $t = 0$. Also, since the voltage of a capacitor cannot change instantaneously,

$$v_C(0^-) = v_C(0^+) = V_0$$



where $v_C(0^-)$ is the voltage across the capacitor just before switching and $v_C(0^+)$ is its voltage immediately after switching. Solving the first-order differential equation

$$\tau \frac{d}{dt} v_C(t) + v_C(t) = v(t)$$ with these conditions gives $$v_C(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \qquad t > 0$$

Thus, $$v_C(t) = \begin{cases} V_0, & t < 0 \\ \underbrace{V_s}_{\text{forced}} + \underbrace{(V_0 - V_s)e^{-t/\tau}}_{\text{natural}}, & t > 0 \end{cases}$$

or $$\boxed{v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]\, e^{-t/\tau}}$$

$v_C(0)$ is the initial voltage at $t = 0^+$ and $v_C(\infty)$ is the final or steady-state value.

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Responses in First Order Systems

We can further find the current through C:

$$i(t) = C\frac{d}{dt}v_C(t) = -\frac{C}{\tau}(V_0 - V_s)e^{-t/\tau} = \frac{V_s - V_0}{R}e^{-t/\tau}$$

and the voltage across R:

$$v_R(t) = R\,i(t) = (V_s - V_0)e^{-t/\tau}$$

We can verify that

$$v_C(t) + v_R(t) = V_s + (V_0 - V_s)e^{-t/\tau} + (V_s - V_0)e^{-t/\tau} = V_s$$

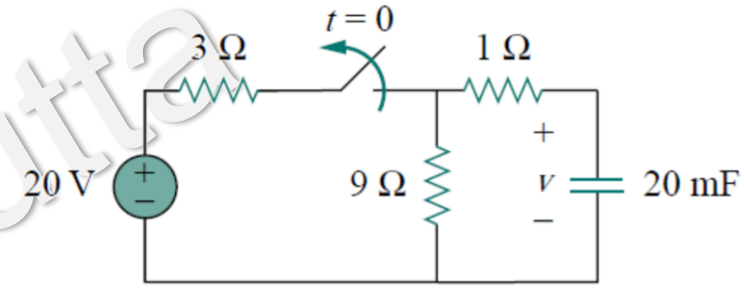
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Responses in First Order Systems

Q: The switch in the circuit has been closed for a long time, and it is opened at $t = 0$. Find $v(t)$ for $t \geq 0$. Calculate the initial energy stored in the capacitor.

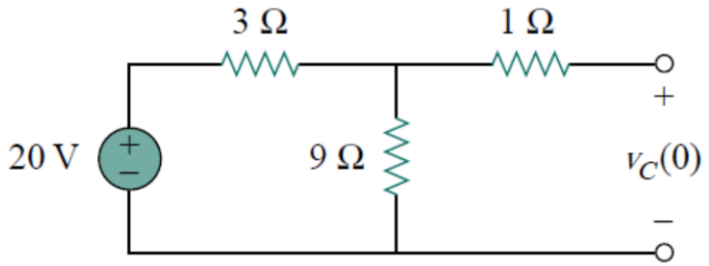
For $t < 0$, the switch is closed; the capacitor is an open circuit to dc. Using voltage division

$$v_C(t) = \frac{9}{9+3}(20) = 15 \text{ V}, \qquad t < 0$$



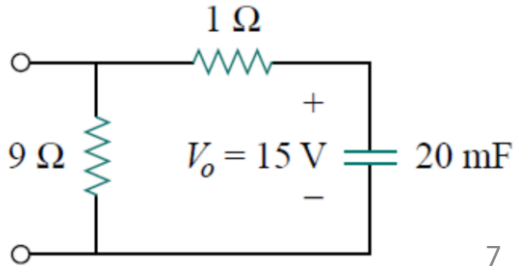
Since the voltage across a capacitor cannot change instantaneously, the voltage across the capacitor at $t = 0^-$ is the same at $t = 0$, or

$$v_C(0) = V_0 = 15 \text{ V}$$



For $t > 0$, the switch is opened, making the *RC* circuit source free. The 1-Ω and 9-Ω resistors in series give

$$R_{\text{eq}} = 1 + 9 = 10 \ \Omega$$



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Responses in First Order Systems

The time constant is

$$\tau = R_{eq}C = 10 \times 20 \times 10^{-3} = 0.2 \text{ s}$$

Thus, the voltage across the capacitor for $t \geq 0$ is

$$v(t) = v_C(0)e^{-t/\tau} = 15e^{-t/0.2} \text{ V}$$

or

$$v(t) = 15e^{-5t} \text{ V}$$

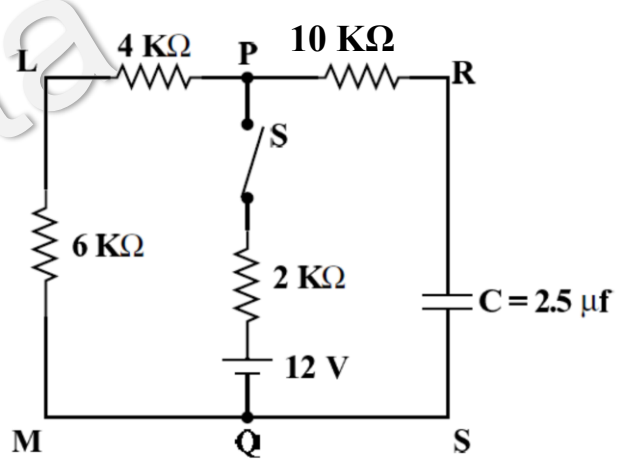
The initial energy stored in the capacitor is

$$w_C(0) = \frac{1}{2}Cv_C^2(0) = \frac{1}{2} \times 20 \times 10^{-3} \times 15^2 = 2.25 \text{ J}$$

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Responses in First Order Systems

Q: In the circuit the capacitor is fully charged when the switch is closed. Calculate the voltage across the fully charged capacitor. Also calculate the voltage across the capacitor and the current in the capacitor circuit 0.05 seconds after opening of the switch.



When the capacitor is fully charged, no current will flow through the capacitor. The capacitor will be charged to a voltage equal to V_{PQ}. However, current, I flowing through the loop PLMQP will be

$$I = \frac{12}{(2+4+6)\times 10^3}\ \text{A}$$

Voltage across terminals P and Q, i.e., V_{PQ} is
V_{PQ} = IR drops across 4 kΩ and 6 kΩ resistors

$$V_{PQ} = \frac{12\times(4+6)\times 10^3}{12\times 10^3}\ \text{V}$$

$$= 10\ \text{V}$$

$$V_{PQ} = v_c = 10\ \text{V}$$

Now, when the switch is opened, at $t = 0$, $v_c = 10$ V. The capacitor will be getting discharged through the resistors 10 kΩ, 4 kΩ, and 6 kΩ in the loop RPLMQSR. The time constant of the circuit, $\tau = RC$.

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Responses in First Order Systems

$$\tau = RC = (10+4+6)\times 10^3 \times 2.5\times 10^{-6} = 20\times 10^3 \times 2.5\times 10^{-6}$$
$$= 0.05 \text{ seconds}$$

Let the capacitor voltage at $t = 0.05$ sec be v_c' and discharging current at $t = 0.05$ be i'.

$$v_c' = v_c e^{-t/\tau} = 10\, e^{\frac{-0.05}{0.05}} = 10\times e^{-1} = 10\times 0.368 = 3.68 V$$

$$I(0) = \frac{v_c}{R_{\text{eq}}} = \frac{10}{20\times 10^3} e^{-t/\tau} = 0.5 \text{ mA}\, e^{-t/\tau}$$

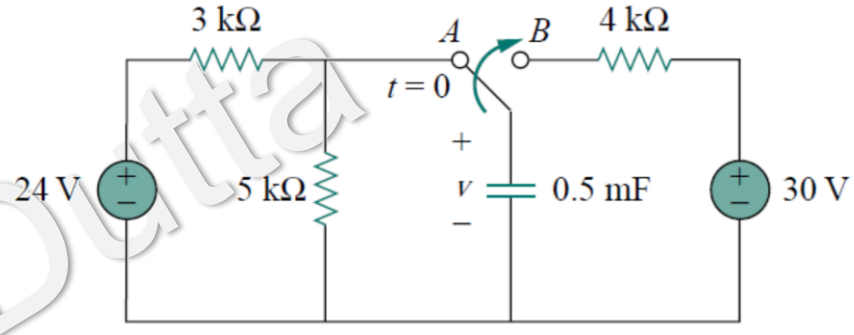
So at $t = 0.05$ s:

$$i' = 0.5 \text{ mA} \cdot e^{-1} \approx 0.184 \text{ mA}.$$

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Responses in First Order Systems

Q: The switch has been in position A for a long time. At $t = 0$, the switch moves to B. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ s and 4 s.



For $t < 0$, the switch is at position A. Since v is the same as the voltage across the 5-kΩ resistor, the voltage across the capacitor just before $t = 0$ is obtained by voltage division as

$$v(0^-) = \frac{5}{5+3}(24) = 15 \text{ V}$$

Using the fact that the capacitor voltage cannot change instantaneously,

$$v(0) = v(0^-) = v(0^+) = 15 \text{ V}$$

For $t > 0$, the switch is in position B. The Thevenin resistance connected to the capacitor is $R_{\text{Th}} = 4$ kΩ, and the time constant is

$$\tau = R_{\text{Th}}C = 4 \times 10^3 \times 0.5 \times 10^{-3} = 2 \text{ s}$$

Since the capacitor acts like an open circuit to dc at steady state, $v(\infty) = 30$ V. Thus,

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} = 30 + (15 - 30)e^{-t/2} = (30 - 15e^{-0.5t}) \text{ V}$$

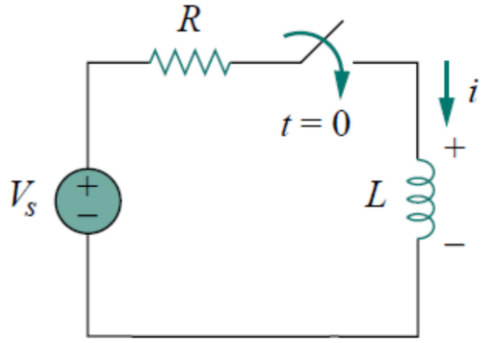
For $t = 1$, $v(1) = 30 - 15e^{-0.5} = 20.902$ V For $t = 4$, $v(4) = 30 - 15e^{-2} = 27.97$ V

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Responses in First Order Systems

- Similarly, for the RL circuit we have:

$$v_R(t) + v_L(t) = R\,i(t) + L\frac{d}{dt}i(t) = v(t) \quad \text{or} \quad \tau\frac{d}{dt}i(t) + i(t) = \frac{v(t)}{R}$$

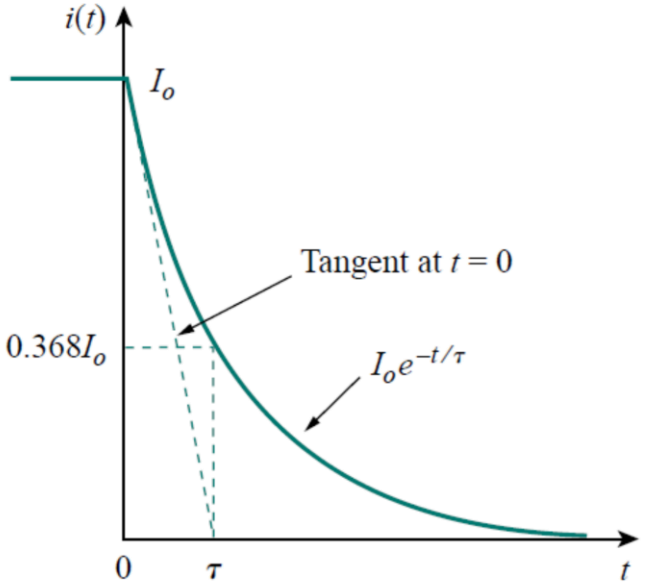


where $\tau = L/R$ is the time constant with the dimension of time:

$$\frac{[L]}{[R]} = \frac{[V][T]}{[I]}\frac{1}{[R]} = \frac{[R][T]}{[R]} = [T]$$

At $t = 0$, we assume that the inductor has an initial current I_0, or $i(0) = I_0$

At $t = 0$, energy stored by inductor $= \frac{1}{2}LI_0^2$



When input is zero, that is, $v(t) = 0$, solving the DE gives the **natural/transient response** of the RL circuit

$$i(t) = I_0 e^{-Rt/L} = I_0 e^{-t/\tau}$$

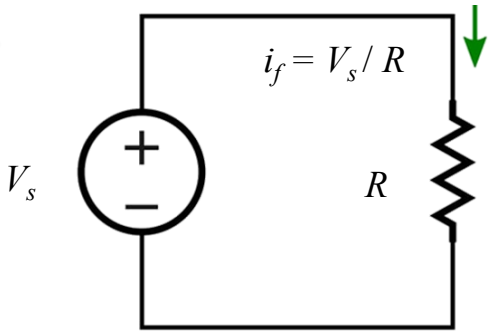
Therefore, the voltage across the resistor is $v_R(t) = iR = I_0 R e^{-t/\tau}$

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Responses in First Order Systems

The **forced response** is the value of the current a long time after the switch is closed. We know that the natural response eventually dies out. At that time, the inductor becomes a short circuit, and the voltage across it is zero. The entire source voltage V_s (say) appears across R. Thus, the forced response is

$$i_f = \frac{V_s}{R}$$



Since the current through the inductor cannot change instantaneously, $i(0^+) = i(0^-) = I_0$

Solving the first-order differential equation $\tau \frac{d}{dt} i(t) + i(t) = \frac{v(t)}{R}$ with the new conditions gives

the **total response** of the RL circuit

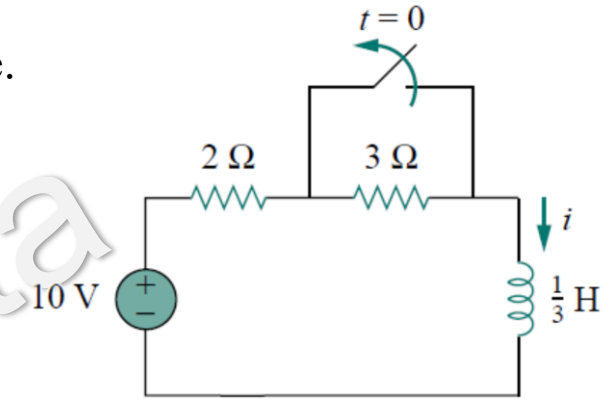
$$i(t) = \underbrace{\frac{V_s}{R}}_{\text{forced}} + \underbrace{\left(I_0 - \frac{V_s}{R}\right) e^{-t/\tau}}_{\text{natural}} \quad \text{or} \quad i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$$

where $i(0)$ and $i(\infty)$ are the initial and final values of i.

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Responses in First Order Systems

Q: Find $i(t)$ in the circuit for $t > 0$. Assume that the switch has been closed for a long time.



When $t < 0$, the 3-Ω resistor is short-circuited, and the inductor acts like a short circuit. The current through the inductor at $t = 0^-$ (i.e., just before $t = 0$) is

$$i(0^-) = \frac{10}{2} = 5 \text{ A}$$

Since the inductor current cannot change instantaneously,

$$i(0) = i(0^+) = i(0^-) = 5 \text{ A}$$

When $t > 0$, the switch is open. The 2-Ω and 3-Ω resistors are in series, so that

$$i(\infty) = \frac{10}{2+3} = 2 \text{ A}$$

The Thevenin resistance across the inductor terminals is

$$R_{\text{Th}} = 2 + 3 = 5 \ \Omega$$

For the time constant,

$$\tau = \frac{L}{R_{\text{Th}}} = \frac{\frac{1}{3}}{5} = \frac{1}{15} \text{ s}$$

Thus,

$$\begin{aligned} i(t) &= i(\infty) + [i(0) - i(\infty)]e^{-t/\tau} \\ &= 2 + (5-2)e^{-15t} = 2 + 3e^{-15t} \text{ A}, \qquad t > 0 \end{aligned}$$

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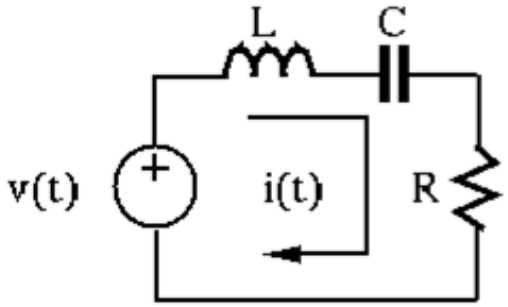
Responses in Second Order Systems

The RCL series circuit here with an input $v(t)$ and output $i(t)$ is described by the following equation:

$$v_R(t) + v_L(t) + v_C(t) = R\,i(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)dt = v_s(t)$$

Taking derivative and dividing by L on both sides we get a 2nd-order linear constant coefficient differential equation (LCCDE):

$$i''(t) + \frac{R}{L}\,i'(t) + \frac{1}{LC}\,i(t) = \frac{1}{L}v_s'(t)$$



Alternatively, as $i(t) = C dv_C(t)/dt$, we have

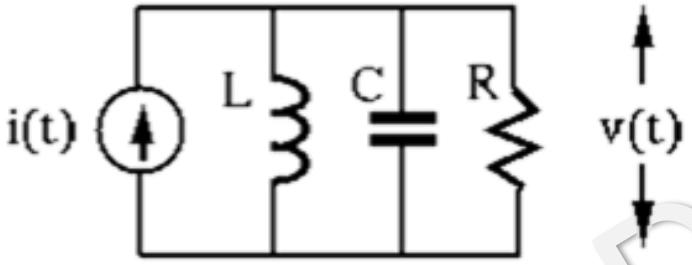
$$v_R(t) = R\,i(t) = RC\frac{d\,v_C(t)}{dt}, \qquad v_L(t) = L\frac{d}{dt}i(t) = L\frac{d}{dt}\left[C\frac{d\,v_C(t)}{dt}\right] = LC\frac{d^2\,v_C(t)}{dt^2}$$

the equation above can be written as a 2nd order ODE in terms of $v_C(t)$:

$$v_C''(t) + \frac{R}{L}v_C'(t) + \frac{1}{LC}v_C(t) = \frac{1}{LC}v_s(t)$$

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Responses in Second Order Systems



The RCL parallel circuit here with input $i(t)$ and output $v(t)$ is described by the following equation:

$$i_R(t) + i_C(t) + i_L(t) = \frac{v(t)}{R} + C\frac{dv(t)}{dt} + \frac{1}{L}\int v(t)\,dt = i_s(t)$$

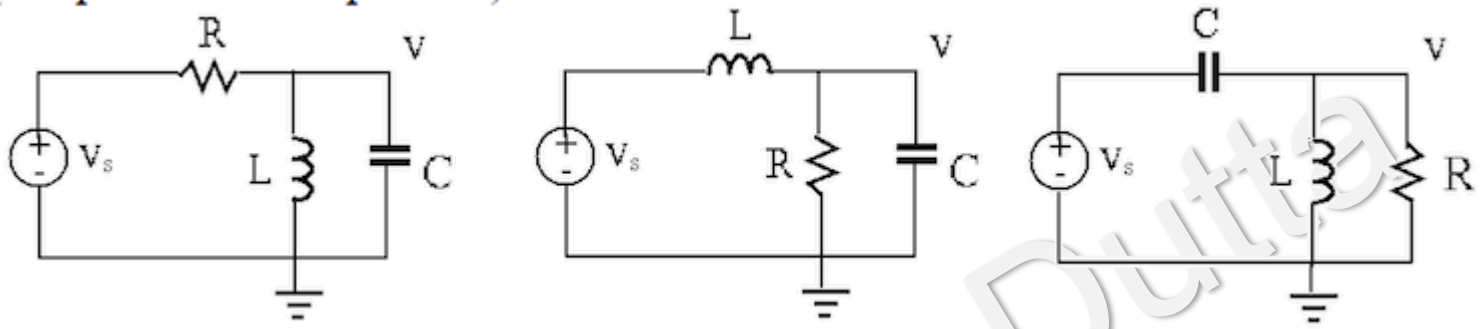
Taking derivative and dividing by C on both sides we get a 2nd-order LCCDE:

$$v''(t) + \frac{1}{RC}\,v'(t) + \frac{1}{LC}\,v(t) = \frac{1}{C}i_s'(t)$$

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Responses in Second Order Systems

- Other RCL circuits (not pure series or parallel):



$$\frac{v_s - v}{R} = C\frac{dv}{dt} + \frac{1}{L}\int v\,dt, \qquad \frac{1}{L}\int (v_s - v)dt = \frac{v}{R} + C\frac{dv}{dt}, \qquad C\frac{d}{dt}(v_s - v) = \frac{v}{R} + \frac{1}{L}\int v\,dt$$

$$v'' + \frac{1}{RC}v' + \frac{1}{LC}v = \frac{v_s'}{RC}, \qquad v'' + \frac{1}{RC}v' + \frac{1}{LC}v = \frac{1}{LC}v_s, \qquad v'' + \frac{1}{RC}v' + \frac{1}{LC}v = v_s''$$

The dimensionality of the coefficient of the first order term is frequency:

$$\left[\frac{R}{L}\right] = \frac{[V][/[I]}{[VT]/[I]} = \frac{1}{[T]}, \qquad \left[\frac{1}{RC}\right] = \frac{[I]}{[V]}\frac{[V]}{[I][T]} = \frac{1}{[T]}$$

The dimensionality of the coefficient of the constant terms is frequency squared: $\left[\frac{1}{LC}\right] = \frac{1}{[VT]/[I]\ [IT]/[V]} = \frac{1}{[T]^2}$

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